Unit

9

Relating the properties of substances to structures and bonding

Unit Key Concepts

- Giant ionic structures
- Giant covalent structures
- Simple molecular structures
- Giant metallic structures

9.1
Comparing the properties of substances with different structures.

Video
Comparing the properties of substances with different structures

Pre-study
A pre-study worksheet on this unit is available for download from *Jing Kung Chemistry Website*.

9.1 A look at the structures of some substances

The structure of a substance is the way in which its particles are arranged. The properties of a substance depend on its structure.

Table 9.1 lists the melting points and electrical conductivities of four common solids: common salt (sodium chloride), quartz, candle wax and copper (Figs. 9.1a–d).

Table 9.1

The melting points and electrical conductivities of four common solids

Name	Melting point	Electrical conductivity
Common salt	high	good conductor when molten or dissolved in water
Quartz	high	non-conductor
Candle wax	low	non-conductor
Copper	high	conductor



Fig. 9.1a Common salt



Fig. 9.1b Quartz



Fig. 9.1c Candle wax

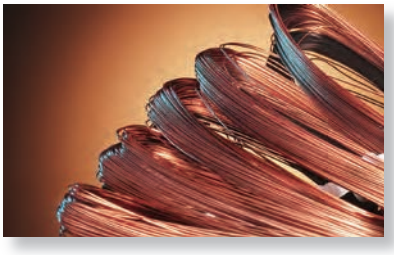


Fig. 9.1d Copper

N1 Refer to page T41.

The four solids shown in Table 9.1 have different melting points and electrical conductivities. This is because they have different structures. N1

- Sodium chloride (common salt) — **giant ionic structure**
- Quartz — **giant covalent structure**
- Candle wax — **simple molecular structure**
- Copper — **giant metallic structure**

AL 2005 Paper 1 Q1(a)(iii)

9.2 Giant ionic structures

Animation
Giant ionic structure — sodium chloride

Ionic compounds such as sodium chloride consist of oppositely charged ions packed closely together in a regular pattern. Strong ionic bonds between oppositely charged ions hold the ions together.

Sodium chloride is made up of sodium ions and chloride ions packed in a regular pattern (Fig. 9.2). This is called a **lattice**.

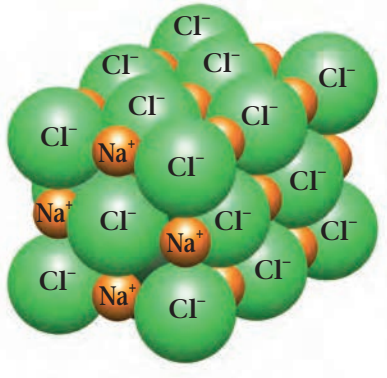


Fig. 9.2 Structure of sodium chloride

In the lattice, each sodium ion is surrounded by 6 chloride ions and each chloride ion is surrounded by 6 sodium ions (Figs. 9.3a–b).



Fig. 9.3a Each sodium ion is surrounded by 6 chloride ions



Fig. 9.3b Each chloride ion is surrounded by 6 sodium ions

giant ionic structure 巨型離子結構　giant covalent structure 巨型共價結構　simple molecular structure 簡單分子結構
giant metallic structure 巨型金屬結構　lattice 晶格

We can also show the structure of sodium chloride by using a ball-and-stick model (Fig. 9.4).

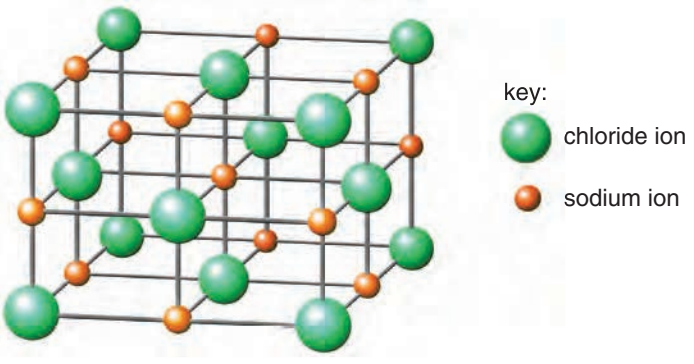


Fig. 9.4 Sodium ions and chloride ions are arranged in a cubic pattern

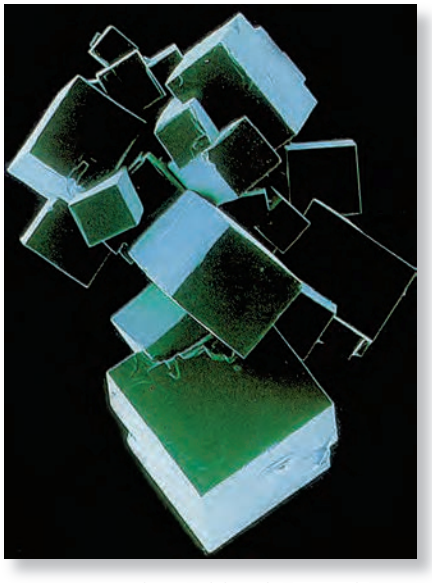


Fig. 9.5 Sodium chloride crystals (magnified)

The ions are arranged in a cubic pattern. This is why sodium chloride crystals are cubic in shape (Fig. 9.5).

Ions in other ionic compounds may be arranged in a different way. Fig. 9.6 shows the structure of caesium chloride (CsCl). The structure consists of a cubic lattice of caesium ions, interpenetrated by another cubic lattice of chloride ions. The eight chloride ions lie at the corners of a cube (green) which is placed in the centre of the caesium structure.

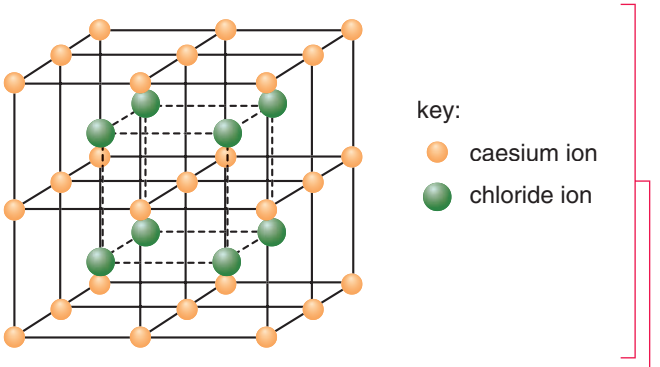


Fig. 9.6 Structure of caesium chloride (CsCl)

ASL 2005 Q2(b)(i)

N2 Refer to page T42.

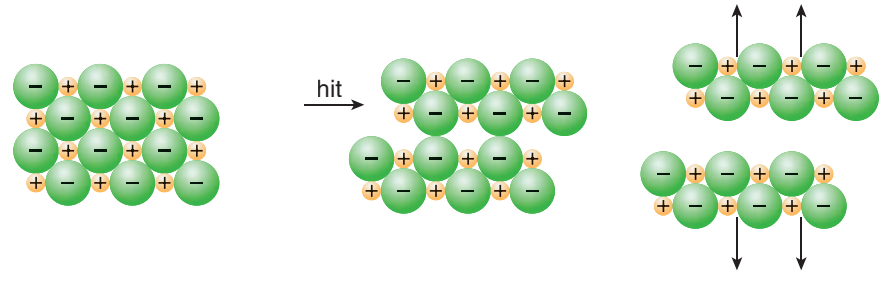
9.3 Properties of ionic compounds N2

Hardness and brittleness

AL 2009 Paper 2 Q3(c)(i)

Ionic compounds are hard. This is due to the strong ionic bonds that hold the ions together. Relative motion of the ions is restricted.

Ionic compounds are **brittle**. They may split apart when hit in the same direction as the layers of ions. This is because they form a lattice of alternating positive and negative ions (Fig. 9.7a). A hit in the direction shown in the diagram may move the ions so that ions with the same charge come together (Fig. 9.7b). The repulsions between thousands of ions in the layers cause the lattice to split (Fig. 9.7c).



(a) Ionic compounds form a lattice of alternating positive and negative ions

(b) A hit may move the ions so that ions with the same charge come together

(c) The repulsions between thousands of ions in the layers cause the lattice to split

Fig. 9.7 The brittleness of ionic compounds

Melting point and boiling point

In order to melt or boil ionic compounds, many of the strong attractive forces (ionic bonds) between the oppositely charged ions have to be overcome. This takes a lot of heat and so ionic compounds have high melting and boiling points (Table 9.2). All ionic compounds are solids at room temperature and pressure.

CE 2005 Paper 2 Q9 & 27
CE 2009 Paper 1 Q9

Table 9.2

Melting points and boiling points of NaCl and MgO		
Ionic compound	Sodium chloride	Magnesium oxide
Chemical formula	NaCl	MgO
Melting point (°C)	808	2 852
Boiling point (°C)	1 465	3 600
State at room temperature and pressure	solid	solid

Do you know ?

Magnesium oxide

Magnesium oxide is used extensively in the construction industry, both in making cement and in making fire-proof construction materials — the fact that it has a melting point of over 2 800 °C makes it ideal for this use.



Fig. 9.8 MgO is used in making fire-proof construction materials

brittle 易碎的

The charges on the Mg^{2+} and O^{2-} ions are greater than those on the Na^{+} and Cl^{-} ions. Furthermore, ions in magnesium oxide are smaller than those in sodium chloride. The ions can pack closer in magnesium oxide. Thus, the electrostatic forces of attraction between ions in magnesium oxide is stronger than that in sodium chloride. The melting point of magnesium oxide is higher than that of sodium chloride.

Solubility

N3 This statement has been modified to a simplified version because the polarity of a solvent has not been covered yet. Actually the statement should read 'many ionic compounds are soluble in water (a polar solvent) but insoluble in non-polar solvents'.

Many ionic compounds are soluble in water but insoluble in **non-aqueous solvents.** N3

In a water molecule, the nucleus of the oxygen atom has a greater attraction for the bonding electrons. This makes the oxygen atom of the molecule slightly negative. It leaves the hydrogen atoms slightly short of electrons and thus slightly positive (Fig. 9.9).

Animation
Dissolving sodium chloride (NaCl) in water

We show the small amount of electrical charge by δ– and δ+ where δ means 'a small amount of'.

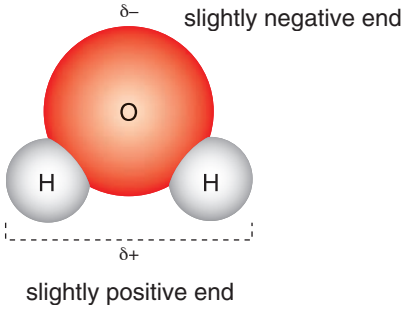


Fig. 9.9 A water molecule with a slightly negative end and a slightly positive end

Reference Website
Refer to an animation showing the dissolving of sodium chloride in water:
http://group.chem.iastate.edu/Greenbowe/sections/projectfolder/flashfiles/thermochem/solutionSalt.html
(accessed July 2014)

The reason why sodium chloride dissolves in water is that water molecules have a strong attraction for ions — the negative ends of water molecules attach to the sodium ions while the positive ends attach to the chloride ions (Fig. 9.10). This attraction causes the sodium ions and chloride ions to move away from the solid and go into the water.

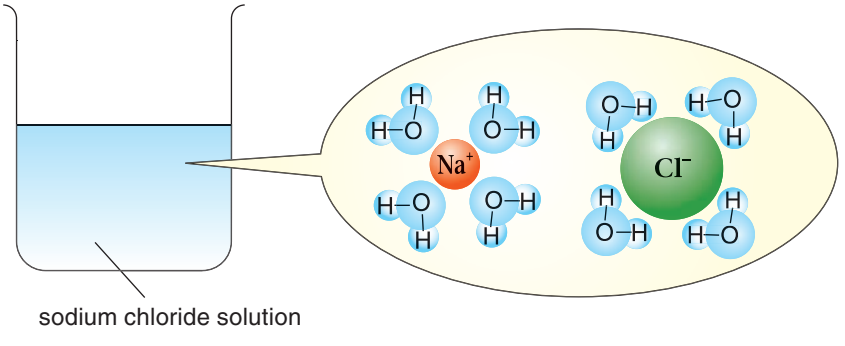


Fig. 9.10 Water molecules attach to sodium ions and chloride ions in a sodium chloride solution

non-aqueous solvent 非水溶劑

There is no such attraction between the ions in sodium chloride and molecules of non-aqueous solvents. Hence sodium chloride is insoluble in non-aqueous solvents.

Magnesium oxide is insoluble in water because the attractive forces between ions in the solid are stronger than those between the ions and water molecules.

Electrical conductivity

CE 2008 Paper 1 Q1(c)
CE 2009 Paper 1 Q9
CE 2010 Paper 2 Q17
DSE 2014 Paper 1A Q20

For any substance to conduct electricity, there must be **mobile** (free moving) charged particles to carry the electric current.

Ionic compounds conduct electricity when molten or dissolved in water◀ but do not conduct electricity in the solid state. When an ionic compound is melted or dissolved in water, the strong ionic bonds are overcome and the ions are free to move (Fig. 9.11).

◀ Ionic compounds are decomposed by electricity during conduction.

Ionic compounds cannot conduct electricity in the solid state as the ions are in fixed positions and not free to move. N4

N4 Examination questions often ask students to explain why an ionic compound is an insulator of electricity in the solid state, but it conducts electricity in the molten state.

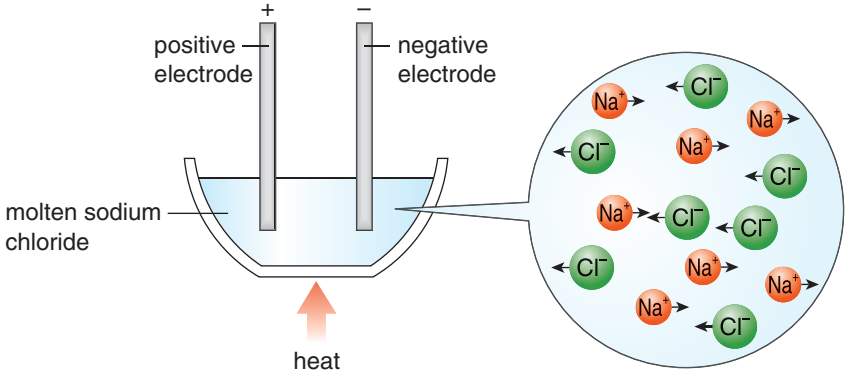


Fig. 9.11 Sodium chloride conducts electricity when molten

Practice 9.1

The electrical conductivity of four substances are listed below.

Substance	Electrical conductivity	
	in solid state	in molten state
W	conducting	conducting
X	conducting	not conducting
Y	not conducting	conducting
Z	not conducting	not conducting

Which of the substances has a giant ionic structure? Explain your answer.

Substance Y has a giant ionic structure.
It cannot conduct electricity in the solid state as the ions are fixed positions and not free to move.
The ions are free to move when Y is in molten state. Thus, it can conduct electricity.

mobile 可自由流動的

Animation
Giant covalent structure — diamond

9.4 Giant covalent structures

Covalent substances are not always made up of small molecules. In some structures, the covalent bonds extend throughout the whole structure. We call these giant covalent structures. **Diamond**, **graphite** and **quartz** are common examples.

The structure of diamond



Fig. 9.12 Diamond is a form of carbon

Diamond is a form of carbon (Fig. 9.12). In diamond, each carbon atom forms four covalent bonds with other carbon atoms. The carbon atoms are tetrahedrally arranged around each other (Fig. 9.13). The network of carbon atoms extends throughout the whole structure, resulting in a giant covalent structure. N5

The structure of silicon is similar to that of diamond.

N5 Silicon and carbon belong to the same group. They have similar structures.



Fig. 9.13 Part of the structure of diamond

The structure of graphite



Fig. 9.14 Graphite is another form of carbon

Graphite is another form of carbon (Fig. 9.14). It differs from diamond only in the arrangement of atoms. Graphite and diamond are **allotropes**.

✓ Allotropes are two (or more) forms of the same element in which the atoms or molecules are arranged in different ways.

AL 2006 Paper 1 Q2(a)(i)

diamond 鑽石　graphite 石墨　quartz 石英　allotrope 同素異形體

In graphite, the carbon atoms are arranged in flat parallel layers as shown in Fig. 9.15.

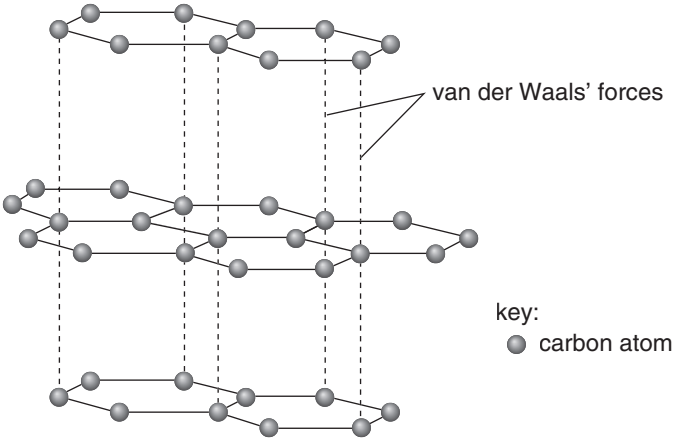


Fig. 9.15 The structure of graphite

AL 2006 Paper 1 Q2(a)(ii)

DSE 2014 Paper 1B Q1(a)(i)
AL 2006 Paper 1 Q2(a)(ii)
AL 2012 Paper 2 Q8(a)

Within each layer, each carbon atom is covalently bonded to three other atoms, forming a hexagonal arrangement. We can consider each layer to be a two-dimensional network of carbon atoms. There are weak **van der Waals' forces** between the adjacent layers in graphite.

Each carbon atom has four outermost shell electrons. Each carbon atom in graphite uses three electrons in forming covalent bonds. The remaining electron is delocalized along the layers of carbon atoms.

CE 2011 Paper 1 Q3(a)(ii)
DSE 2014 Paper 1B Q1(a)(ii)
AL 2008 Paper 2 Q4(c)(ii)

N6 Quartz has a giant covalent structure, NOT 'giant covalent bond'. ✘

The structure of quartz



Fig. 9.16 Quartz

There are several forms of silicon dioxide. The silicon dioxide found in the mineral quartz (Fig. 9.16) has a structure similar to that of N6 diamond.

DSE 2013 Paper 1A Q1

CE 2006 Paper 1 Q5(a)

In a quartz crystal, each silicon atom is joined to four oxygen atoms by covalent bonds, while each oxygen atom is joined to two silicon atoms by covalent bonds. Hence the giant structure consists of a network of covalent bonds (Fig. 9.17). The overall ratio of silicon to oxygen is 1 : 2. Hence the chemical formula of silicon dioxide is SiO_2.

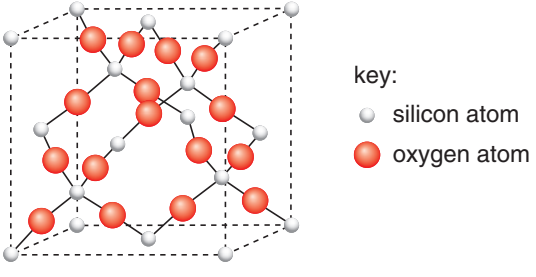


Fig. 9.17 The structure of quartz

van der Waals' forces 范德華力

9.5 Properties of substances with giant covalent structures

N7 Refer to page T42.

Table 9.3 summarizes some properties of quartz, diamond and graphite. N7

Table 9.3

Some properties of quartz, diamond and graphite

Substance	State at room temperature and pressure	Hardness	Melting point (°C)	Solubility in water	Electrical conductivity
Quartz	solid	very hard	1 610	insoluble	non-conductor
Diamond	solid	very hard	3 500	insoluble	non-conductor
Graphite	solid	soft	3 730	insoluble	conductor

Hardness

Substances with giant covalent structures are hard (except graphite). These substances consist of a network of covalent bonds. Relative motion of the atoms is restricted. Diamond is the hardest substance known and is used in cutting tools (Fig. 9.18). N8

CE 2011 Paper 1 Q3(c)
DSE 2013 Paper 1A Q1



Fig. 9.18 Diamond is so hard that it can be used to cut stones. This cutting wheel is edged with diamond

N8 Silicon and carbon react to form silicon carbide (SiC). The crystal structure of silicon carbide is similar to that of diamond. Silicon carbide is very hard.

N9 SiO_2 is very stable in moist air.

Melting point

To melt a substance with a giant covalent structure, a lot of heat is needed to break the strong covalent bonds between the atoms. Thus, these substances have high melting points.

DSE 2014 Paper 1B Q1(b)

Solubility

Substances with giant covalent structures are insoluble in water and non-aqueous solvents. N9 This is because the atoms are held together by strong covalent bonds and it is very difficult to separate the atoms.

Electrical conductivity

Substances with giant covalent structures do not conduct electricity (except graphite). This is because all the outermost shell electrons are either held by individual atoms or involved in covalent bonding. There are no mobile electrons or ions.

AL 2008 Paper 2 Q4(c)(ii)

9.6 Applications of graphite

The properties of a substance determine its use. For example, graphite is widely used because of its unique combination of properties.

Graphite is soft and easily scratched. The forces between the layers of carbon atoms are weak. The layers can easily slide past each other when a force is applied. The layers also flake off readily.

As a lubricant

One of the main uses for graphite — as a **lubricant** (Fig. 9.19) — results from the sliding of one layer past another within the structure.



Fig. 9.19 A drop of lubricant oil containing graphite

Lead pencils

CE 2011 Paper 1 Q3(a)(i)

The 'lead' in pencils is actually graphite (Fig. 9.20). The mark it leaves on the paper is the result of layers of carbon atoms flaking off the structure.



Fig. 9.20 Writing with a lump of graphite

Lead metal can also be used to draw markings. However, using lead to make pencil cores is not as good as using graphite. The reason is that strong metallic bonding exists in lead and lead tears off less readily than graphite does.



Fig. 9.21 The electrode of this cell is made of graphite

Electrical applications

Graphite is a good conductor of electricity. When a voltage is applied, the delocalized electrons can move along the layers.

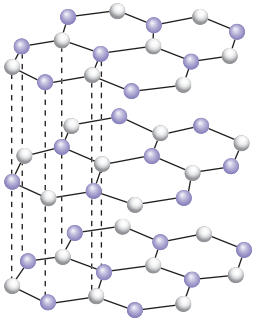
Graphite is used as electrodes in dry cells (Fig. 9.21) and in the manufacture of carbon brushes in electric motors.

lubricant 潤滑劑

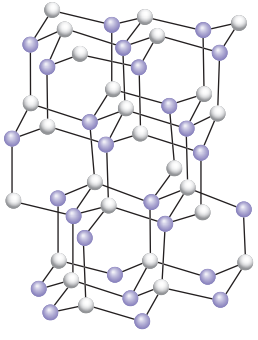
Practice 9.2

Boron nitride, BN, exists in two different forms. The forms have structures that are similar to graphite and diamond.

Look at the structures of these two forms of boron nitride.



structure A



structure B

a) Explain why boron nitride with structure A is slippery.

b) It is claimed that boron nitride with structure B is as hard as diamond.

i) Explain why this form of boron nitride is hard.

ii) Explain whether this form of boron nitride can conduct electricity.

a) The attractive forces between the layers are weak. Thus, the layers can slide over each other easily.
b) i) This form of boron nitride has a giant covalent structure. The strong covalent bonds restrict the relative motions of the atoms.
ii) This form of boron nitride cannot conduct electricity because all the outermost shell electrons are either held by individual atoms or involved in covalent bonding. There are no mobile electrons or ions.

9.7 Simple molecular structures

Animation
Simple molecular structure — iodine

In most non-metals, atoms join together to form small molecules. For example, iodine exists as diatomic molecules (I_2). Most of the compounds of non-metals with other non-metals also have simple molecular structures. This is true to simple compounds such as ammonia and water.

In these substances, strong covalent bonds hold the atoms in each molecule together. Much weaker van der Waals' forces hold the separate molecules together. N10

Van der Waals' forces also hold atoms in a noble gas together.

N10 When considering substances with simple molecular structures (e.g. carbon dioxide and iodine), it is important to distinguish between the covalent bonds holding atoms within the molecules and the weak attractive forces between the molecules.

The structure of iodine

In each iodine molecule (Fig. 9.22), a strong covalent bond holds the two iodine atoms together.



Fig. 9.22 Iodine crystals

CE 2005 Paper 2 Q7

Fig. 9.23 shows the structure of iodine. The iodine molecules are N11 packed close to one another in a regular pattern. Weak van der Waals' forces hold the molecules together. The pattern is repeated millions N12 of times, and the result is a crystal.

N11 Examination questions often ask about the structure of iodine.

N12 When iodine sublimes, it absorbs energy as energy is required to overcome the attractive forces between iodine molecules.

N13 Although carbon and silicon belong to the same group, they form oxides with different structures. In quartz, attractions between the atoms are covalent bonds. In carbon dioxide, attractive forces between the molecules are weak van der Waals' forces.

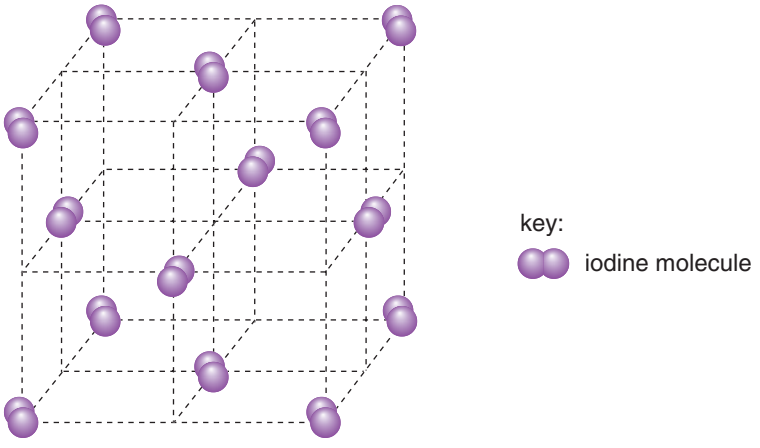


Fig. 9.23 The arrangement of iodine molecules in its crystal

The structure of dry ice



Fig. 9.24 Dry ice

Dry ice (Fig. 9.24) consists of separate carbon dioxide molecules and is used to create a misty effect on the stage (Fig. 9.25).

In each carbon dioxide molecule, strong covalent bonds hold the carbon atoms and oxygen atoms together. The molecules are packed close to one another in a regular pattern. Weak van der Waals' forces N13 hold the molecules together (Fig. 9.26).



Fig. 9.25 Dry ice is used to create a misty effect on the stage

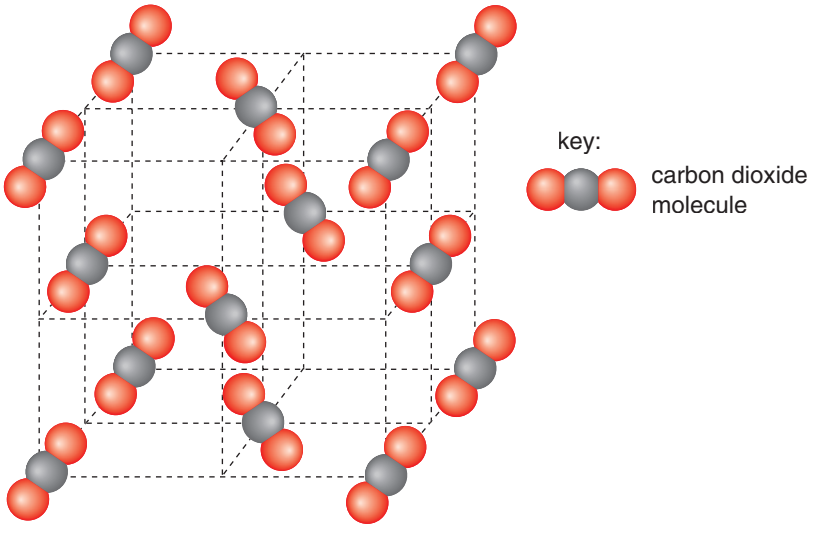


Fig. 9.26 The arrangement of carbon dioxide molecules in dry ice

dry ice 乾冰

N14 Refer to page T43.

9.8 Properties of substances with simple molecular structures N14

Hardness

Solid substances with simple molecular structures are usually soft because the attractive forces between the molecules are weak.

Melting point and boiling point

When simple molecular substances melt or boil, it is the van der Waals' forces between molecules that are overcome. The covalent bonds do not break.

Simple molecular substances have low melting and boiling points because the attractive forces between the molecules are weak. Little heat is needed to separate the molecules. Many simple molecular substances are gases, liquids or solids with low melting points (Table 9.4).

CE 2009 Paper 1 Q9
CE 2011 Paper 2 Q5

Table 9.4

The melting points and boiling points of some common simple molecular substances

Substance	Ethanol	Hydrogen chloride	Iodine	Oxygen	Water
Chemical formula	C_2H_5OH	HCl	I_2	O_2	H_2O
Melting point (°C)	–117	–114	114	–219	0
Boiling point (°C)	78	–85	184	–183	100
State at room temperature and pressure	liquid	gas	solid	gas	liquid

CE 2007 Paper 2 Q32

Solubility

Substances with simple molecular structures are usually slightly soluble N15 or insoluble in water◂ but very soluble in non-aqueous solvents.

The solubility of a simple molecular substance is determined by the following attractive forces:

- attractive forces between solute molecules;
- attractive forces between solvent molecules;
- attractive forces between solute and solvent molecules.

◂ Some substances with simple molecular structures are quite soluble in water, for example, sugar.

N15 As for the solubility of substances with simple molecular structures, it is stated in the text that 'substances with simple molecular structures are usually slightly soluble or insoluble in water but very soluble in non-aqueous solvents'. This statement is again a simplified version. Actually the statement should read 'substances with simple molecular structures are usually slightly soluble or insoluble in polar solvents but very soluble in non-polar solvents'.



Fig. 9.27 Aqueous solution of iodine is pale brown (left), and iodine dissolved in a non-aqueous solvent to give a purple solution (right)

Take iodine as an example. It is slightly soluble in water (Fig. 9.27). The attractive forces between water molecules are quite strong. The weak attractive forces between iodine molecules and water molecules are not strong enough to overcome the attractive forces between the water molecules (Fig. 9.28). Hence iodine molecules and water molecules do not mix easily.

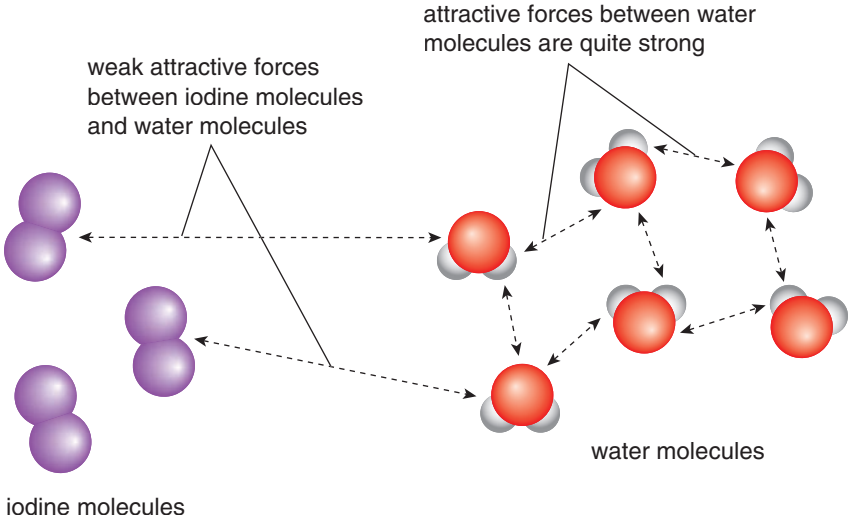


Fig. 9.28 Iodine is slightly soluble in water

On the other hand, iodine is very soluble in non-aqueous solvents (Fig. 9.27). In this case, the attractive forces between the iodine molecules and the solvent molecules are much the same as in the pure substances (Fig. 9.29). Hence iodine molecules and the solvent molecules mix together easily.

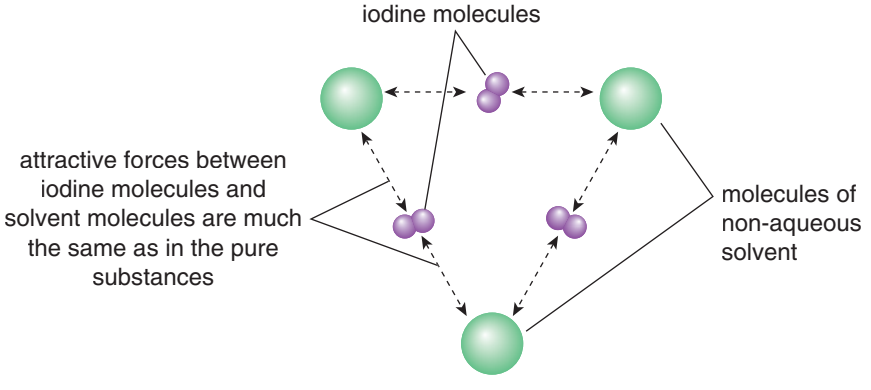


Fig. 9.29 Iodine is very soluble in non-aqueous solvents

N16 Molten sulphur contains mobile sulphur molecules. Molten sulphur does not conduct electricity.

Aqueous solutions of some substances with simple molecular structures conduct electricity. For example, hydrogen chloride reacts with water to form mobile hydrogen ions and chloride ions.

Electrical conductivity

Substances with simple molecular structures do not conduct electricity because they do not contain mobile electrons or ions. **N16**

CE 2007 Paper 2 Q28
CE 2009 Paper 1 Q9

Example 9.1

Q Chlorine and argon belong to the same period of the periodic table.

Explain why the boiling point of chlorine is higher than that of argon.

DSE 2012 Paper 1B Q1(d)

A The boiling point of a substance depends on the strength of its intermolecular attractions.

Chlorine exists as diatomic molecules while argon as monoatomic molecules. A chlorine molecule has a greater size than an argon molecule. Stronger van der Waals' forces exist in chlorine.

Practice 9.3

The table below gives the boiling points of some substances made up of molecules. Arrange these substances in increasing order of the strength of attractive forces between molecules.

Substance	Boiling point (°C)
Ammonia	–33
Ethanol	78
Hydrogen	–101
Water	100

Their boiling points increase with the strength of attractive forces between molecules.
Thus, the increasing order of strength of attractive forces between molecules is:
hydrogen < ammonia < ethanol < water

9.9 Giant metallic structures

Animation
Giant metallic structure

In a metal, the positive metallic ions are packed tightly together in a regular three-dimensional arrangement to form a giant metallic structure (Fig. 9.30). The structure is held together by the attractions between the positive metallic ions and a 'sea' of delocalized electrons.

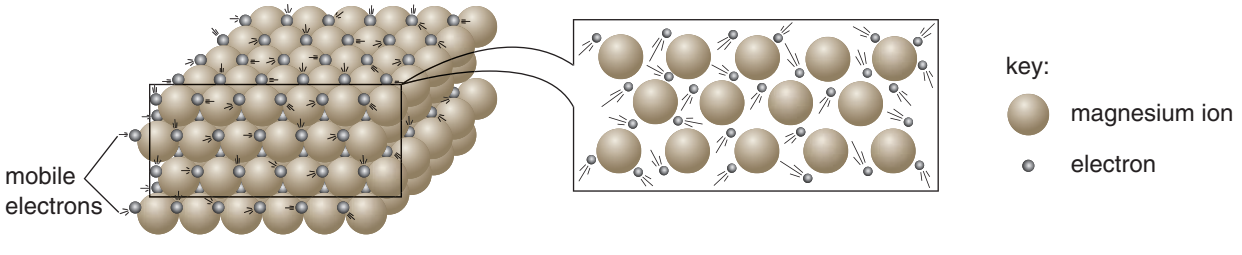


Fig. 9.30 The giant structure of magnesium

9.10 Properties of metals

AL 2007 Paper 1 Q9

Density

Metals generally have high densities due to the close packing of their positive metallic ions (Table 9.5).

Table 9.5

The densities of some common metals	
Metal	**Density (g cm^{-3})**
Copper	8.9
Gold	19.3
Iron	7.9

Melting point

The positive metallic ions in a metal are closely packed with strong attractive forces between these ions and delocalized electrons. To melt a piece of metal, a lot of heat is needed to overcome the strong attractive forces. Hence metals generally have high melting points N17 (Table 9.6).

CE 2009 Paper 1 Q9

N17 Sodium, magnesium and aluminium in Period 3 are metals. The strength of the metallic bond depends on the number of delocalized electrons in the metal structure. Sodium has one outermost shell electron per atom, magnesium has two, while aluminium has three. The strength of metallic bond and hence the melting point increase from sodium to aluminium.

Table 9.6

The melting points of some common metals	
Metal	**Melting point (°C)**
Copper	1 083
Gold	1 063
Iron	1 538

Electrical and heat conductivities

Metals are good conductors of electricity when solid and when molten. This is because their delocalized electrons are mobile. When a metal is connected to a battery, electrons flow towards the positive terminal and the negative terminal can supply electrons into the metal (Fig. 9.31). N18

N18 A metal conducts electricity without decomposition. The conductivity of a metal is a physical property.

CE 2009 Paper 1 Q9

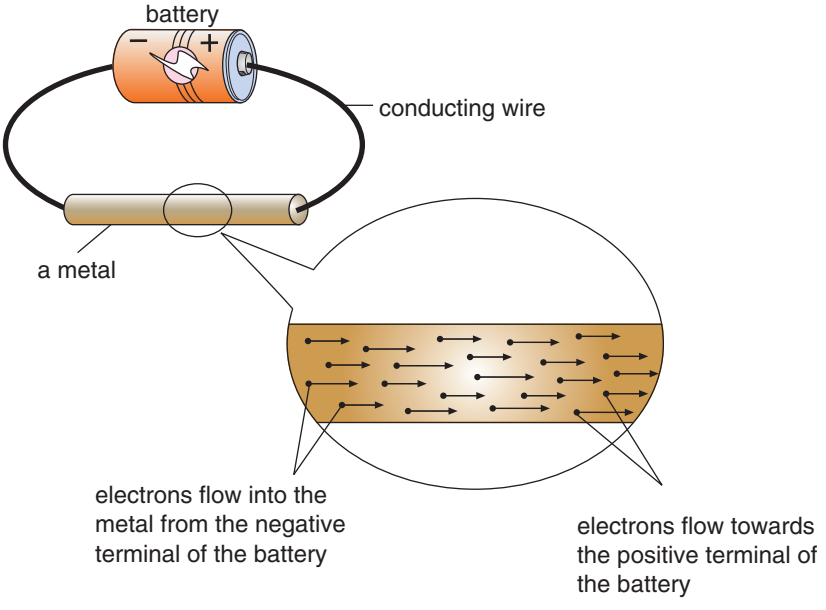


Fig. 9.31 Movement of electrons in a piece of metal when the metal is connected to a battery

Metals are also good conductors of heat. When heating one end of a piece of metal, the mobile electrons get more energy. They move faster and collide with neighbouring electrons. This helps to transfer the heat (Fig. 9.32).

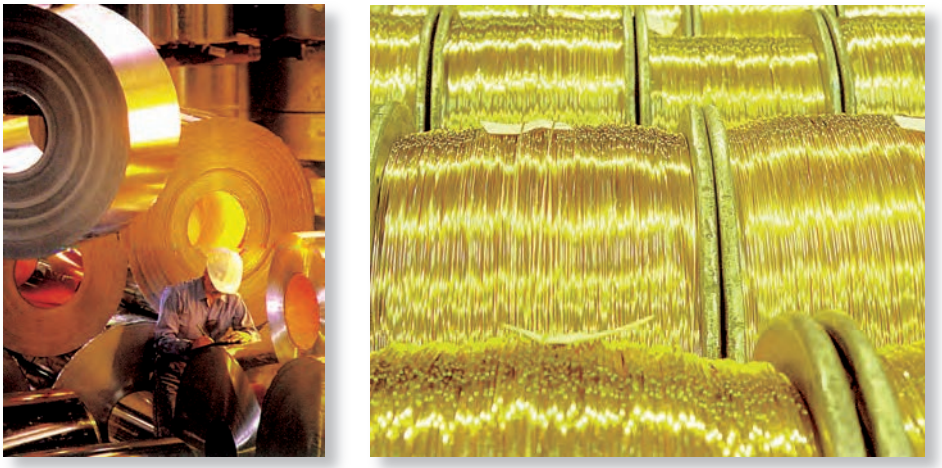


Fig. 9.32 Metals are good conductors of heat due to the movement of mobile electrons in metals

This contrasts with ionic lattices which are hard but brittle. Metals (generally) are hard but malleable and ductile.

Malleability and ductility

Metals are both **malleable** and **ductile**. Metals can be hammered or rolled into thin sheets as they are malleable (Fig. 9.33a). Ductile metals can be drawn into wires (Fig. 9.33b).



(a) (b)

Fig. 9.33 (a) Aluminium can be pressed and rolled into sheets (b) Copper wires rolled into spools

AL 2009 Paper 2 Q3(c)(i)

Positive metallic ions in metals are packed in layers. When we apply a force to a piece of metal, the layers of metallic ions slide over each other to new positions (Fig. 9.34). New metallic bonds are re-formed between the metallic ions in new positions and the delocalized electrons. The delocalized electrons continue to hold the ions together. This explains why metals can be hammered into thin sheets or pulled into long, thin wires. N19

The high strength and hardness of most metals are also due to the strong metallic bonds between the metallic ions and the delocalized electrons.

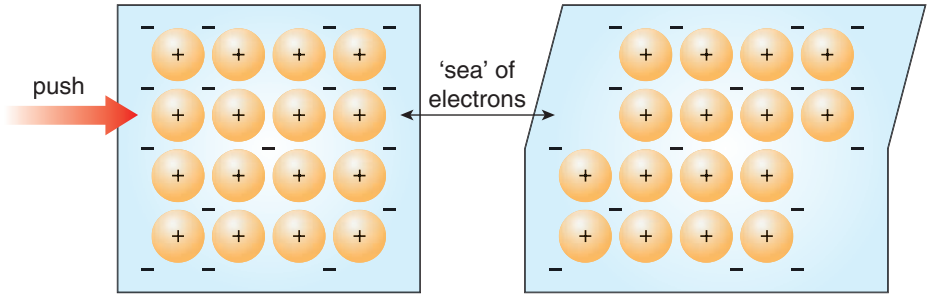


Fig. 9.34 New metallic bonds are re-formed between the metallic ions in new positions and the delocalized electrons

N19 Mild steel is less malleable than iron as carbon atoms cause a distortion to the regular arrangement of metallic ions in iron.

Do you know

Superconductors

At very low temperatures, some metals, including mercury, lead and tin, allow electricity to flow through them with no resistance. These conductors are called *superconductors*. An electric current can flow through a superconductor with no loss of energy. This is called loss-free power transmission.

Superconductors can be used to make super-fast electronic circuits and very powerful electromagnets.

Superconductors also allow for the development of magnetically levitated trains. Such trains (now in service in Shanghai) 'float' above the track using superconducting magnets. Free of friction, these trains glide along at high speeds.

malleable 可展的 ductile 可延的 superconductor 超導體

Practice 9.4

The table below lists the electrical conductivities of two substances in the solid state.

	Lead	Lead(II) bromide
Electrical conductivity in solid state	conducting	non-conducting

a) Describe the structure and bonding in lead.

b) Explain the difference in electrical conductivity between lead and lead(II) bromide in the solid state.

c) Will lead(II) bromide conduct electricity in the molten state? Explain your answer.

Practice 9.4

a) A regular three-dimensional arrangement of positive metallic ions in a 'sea' of electrons.
A metallic bond refers to the strong electrostatic forces of attraction between the positive metallic ions and a 'sea' of delocalized electrons.

b) Solid lead can conduct electricity because its delocalized electrons are mobile.
Solid lead(II) bromide cannot conduct electricity because the ions in it are in fixed positions and not free to move.

c) Lead(II) bromide will conduct electricity in the molten state because ions in the compound become mobile.

9.11 Predicting the properties of a substance from its structure

We have learnt the relationship between bonding, structures and properties of different substances. From the group numbers of elements that make up a compound, we can tell the type of bonding present and the structure of the compound. We can then predict the properties of the compound from its structure.

Example 9.2

Q Strontium (Sr) is a Group II element below calcium in the periodic table. It combines with chlorine to form a compound called strontium chloride.

a) Predict the type of bonding present in strontium chloride. Explain your answer.

b) Draw an electron diagram of strontium chloride (showing electrons in the *outermost shells* only).

c) What type of structure does strontium chloride have?

d) State the chemical formula of strontium chloride.

e) Predict and explain the following properties of strontium chloride:

i) hardness;

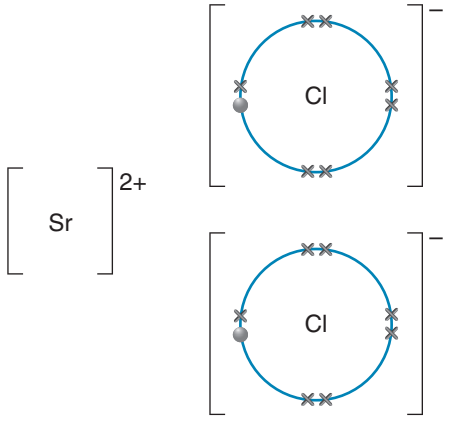
ii) melting point; and

iii) electrical conductivity.

Continued on next page

A a) When strontium (a metal) combines with chlorine (a non-metal), ionic bonding occurs.

b)



c) Strontium chloride has a giant ionic structure.

d) The chemical formula of strontium chloride is $SrCl_2$.

e) i) The compound is hard because the ions in it are held together by strong ionic bonds.

ii) The compound has a high melting point. To melt the compound, many of the strong ionic bonds between the ions have to be overcome. This takes a lot of heat.

iii) The compound conducts electricity when molten or dissolved in water because the ions are free to move.

Example 9.3

Q The following table lists the atomic numbers of three elements, X, Y and Z.

Element	Atomic number
X	6
Y	8
Z	14

a) P is a compound formed from elements X and Y.

i) Predict the type of bonding present in P. Explain your answer.

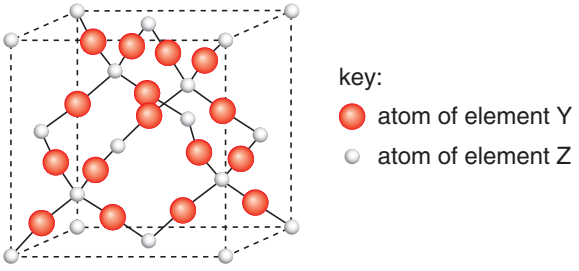
ii) Draw an electron diagram of P (showing electrons in the *outermost shells* only).

iii) What type of structure does P have?

iv) State the chemical formula of P.

Continued on next page

b) Q is a compound formed from elements Y and Z. The structure of Q is shown below:



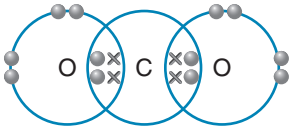
i) What type of structure does Q have?

ii) State the chemical formula of Q.

iii) With reference to its structure, explain why Q can be used as an *abrasive*.

c) State with explanation, which compound (P or Q) has a higher melting point.

A1 a) i) Element X is carbon (a non-metal) while element Y is oxygen (a non-metal). Covalent bonding occurs when they combine.

ii) 

iii) P has a simple molecular structure.

iv) CO_2

b) i) Giant covalent structure

ii) SiO_2

iii) The strong covalent bonds in Q restrict the relative motion of the atoms.

Hence Q is hard and strong.

c) Compound Q has a giant covalent structure while compound P has a simple molecular structure. Weak van der Waals' forces hold the molecules of P together.

To melt Q, a lot of heat is needed to break the strong covalent bonds between the atoms.

In contrast, the attractive forces between molecules of P are weak. Little heat is needed to separate the molecules. Hence Q has a higher melting point than P.

abrasive 磨蝕劑

c) Q has a higher melting point.
Q has a giant ionic structure. To melt Q, many of the strong ionic bonds between the oppositely charged ions have to be overcome. This takes a lot of heat.
P has a simple molecular structure. The attractive forces between molecules of P are weak. Little heat is needed to separate the molecules. Hence Q has a higher melting point than P.

Practice 9.5

Electronic arrangements of atoms of three elements are given below.

Element	Electronic arrangement of atom
X	1
Y	2,8,7
Z	2,8,8,1

a) Compound P is formed from the reaction between elements X and Y.

i) Predict the type of bonding present in P.

ii) Draw an electron diagram of P (showing electrons in the *outermost shells* only).

b) Compound Q is formed from the reaction between elements Y and Z.

i) Predict the type of bonding present in Q.

ii) Draw a diagram to show the structure of Q.

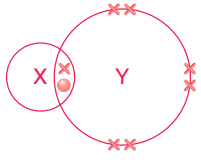
iii) Explain why solid Q is brittle.

c) State, with explanation, which compound (P or Q) has a higher melting point.

Practice 9.5

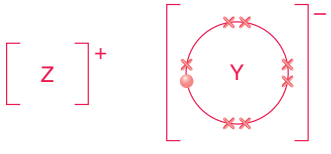
a) i) Covalent bonding

ii)



b) i) Ionic bonding

ii)



iii) The ions in the compound form a lattice of alternating positive and negative ions.
A hit on the lattice may move the ions so that ions with the same charge come together.
The repulsions between thousands of ions cause the lattice to split.

9.12 Predicting the bonding and structure of a substance from its properties

Bonding and electrical conductivity

There are three types of strong bonding — ionic, covalent and metallic. These three types of bonding give rise to different properties.

The property that best tells us what sort of bonding we have is electrical conductivity. Metals conduct electricity well, in both the solid and molten states, due to their metallic bonding. Ionic compounds only conduct electricity when molten or dissolved in water. Generally, covalent substances do not conduct electricity.

Thus, we can decide what type of bonding a substance has by looking at how it conducts electricity (Table 9.7).

ASL 2012 Paper 1 Q11
AL 2012 Paper 1 Q9

Table 9.7

The pattern of electrical conductivities of substances with different types of bonding

Bonding in substance	Electrical conductivity		
	in solid state	in molten state	in aqueous solution
Metallic	✔	✔	does not dissolve but may react
Ionic	✘	✔	✔
Covalent	✘	✘	✘ (but may react)

CE 2009 Paper 2 Q22
CE 2010 Paper 2 Q17

N20 Refer to page T43.

Structure and melting point N20

The property that best tells us if a structure is giant or simple molecular is the melting point (or boiling point). Substances with simple molecular structures have low melting points while those with giant structures have high melting points.

Generally, any substance with a high melting point also has a high boiling point. However, there are some substances, such as iodine, that sublime, i.e. these solid substances turn directly to vapour.

A substance with a low melting point has a simple molecular structure. We know atoms in all simple molecular substances are covalently bonded. So all substances with low melting points must have covalent bonding.

However, a substance with covalent bonding may have a giant structure and therefore may have a high melting point.

Table 9.8 summarizes the melting points of substances with different types of bonding and structures.

Table 9.8

Melting points of substances with different types of bonding and structures

Bonding	Structure	Melting point
Metallic	giant	high
Ionic	giant	high
Covalent	giant	high
	simple molecular	low

CE 2009 Paper 2 Q22

Practice 9.6

The properties of four solids are given below.

Solid	Melting point	Electrical conductivity	Solubility in water
W	high	good	insoluble
X	high	non-conducting	soluble
Y	low	non-conducting	soluble
Z	very high	non-conducting	insoluble

Which of the solids is likely to have

a) a simple molecular structure?
Y

b) a giant ionic structure?
X

c) a giant metallic structure?
W

d) a giant covalent structure?
Z

CE 2007 Paper 2 Q18
ASL 2009 Paper 1 Q2(c)
AL 2009 Paper 1 Q3(b)(iii)

Worksheet

Two related worksheets with answers are available for download from *Jing Kung Chemistry Website*. Worksheet ① consists of additional questions for promoting active reading. (Note: Questions in this worksheet also cover those in the textbook.) Worksheet ②, requiring students to indicate the relationship among various terms by drawing, may help the less able readers improve their reading skills.

Chemistry Magazine

Aluminium — a unique and versatile metal

Aluminium is used extensively and diversely in the modern world due to its many special properties.

Packaging

Aluminium is corrosion resistant and non-toxic. It is also easily shaped and impermeable to gases and water vapour. Hence it is widely used in packaging, including drink cans, bottle tops, foil wrappings and foil containers (Fig. 9.35).



Fig. 9.35 Aluminium is widely used in food packaging

Transport

Low density and high strength make aluminium ideal for the construction of aircraft (Fig. 9.36). Less weight means less fuel consumption.



Fig. 9.36 Aluminium and its alloys are used in the construction of aircraft

Buildings

Aluminium's corrosive resistance and low density make it ideal for use in cladding, windows, door frames (Fig. 9.37) and roofing.



Fig. 9.37 Door frames made of aluminium

Questions

1 Using aluminium as an example, describe the bonding in metals.

2 Aluminium is used to make overhead power transmission lines. List properties of aluminium that make it suitable for this application.

3 Tin-plated steel has been used to make food cans since the invention of canned food. Aluminium is now being used to make cans too.

Suggest ONE advantage and ONE disadvantage of using aluminium to make food cans.

Refer to page T44.

Have you mastered?

Key terms

giant ionic structure 巨型離子結構 133 giant covalent structure 巨型共價結構 133

simple molecular structure 簡單分子結構 133 giant metallic structure 巨型金屬結構 133

lattice 晶格 133 brittle 易碎的 135 non-aqueous solvent 非水溶劑 136

mobile 可自由流動的 137 diamond 鑽石 138 graphite 石墨 138 quartz 石英 138

allotrope 同素異形體 138 van der Waals' forces 范德華力 139 lubricant 潤滑劑 141

dry ice 乾冰 143 malleable 可展的 149 ductile 可延的 149

Checklist

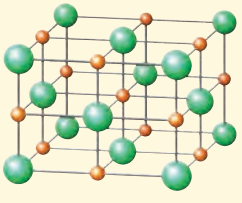
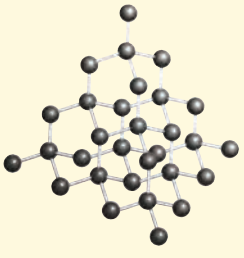
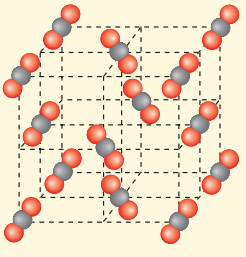
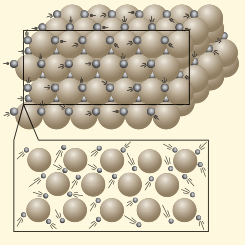
After studying this unit, you should be able to

- ☐ describe giant ionic structures of substances such as sodium chloride and caesium chloride;
- ☐ state and explain the properties of ionic compounds in terms of their bonding and structures;
- ☐ describe giant covalent structures of substances such as diamond, graphite and quartz;
- ☐ state and explain the properties of giant covalent substances in terms of their bonding and structures;
- ☐ describe simple molecular structures of substances such as carbon dioxide and iodine;
- ☐ recognize that van der Waals' forces exist between molecules;
- ☐ state and explain the properties of simple molecular substances in terms of their bonding and structures;
- ☐ describe the general properties of metals;
- ☐ relate the properties of metals to their giant metallic structures;
- ☐ compare the structures and properties of substances with giant ionic, giant covalent, simple molecular and giant metallic structures;
- ☐ deduce the properties of substances from their bonding and structures, and vice versa;
- ☐ explain the applications of substances according to their substances.

(Put a '✔' in the box if you have acquired the knowledge concerned.)

Summary

1 The following table summarizes the properties of substances with the four types of structures: giant ionic, giant covalent, simple molecular and giant metallic.

	Type of structure			
	Giant ionic	**Giant covalent**	**Simple molecular**	**Giant metallic**
Where this type of structure is found	compounds formed between metals and non-metals	Group IV elements and some of their compounds	some non-metallic elements and some compounds formed between non-metals	metals
Examples	sodium chloride, magnesium oxide	diamond, graphite, silicon dioxide	hydrogen, iodine, ammonia, dry ice	aluminium, copper, magnesium
Constituent particles	oppositely charged ions	atoms	small molecules	positive metallic ions surrounded by a 'sea' of delocalized electrons
	example: NaCl 	example: diamond 	example: dry ice 	example: magnesium 
Bonding	strong ionic bonds	atoms joined by strong covalent bonds	strong covalent bonds within the molecules; weak van der Waals' forces hold separate molecules together	strong metallic bonds
Common state(s) at room temperature and pressure	solid	solid	solid, liquid or gas	solid

Have you mastered?

Continued on next page

	Type of structure			
	Giant ionic	**Giant covalent**	**Simple molecular**	**Giant metallic**
Hardness	hard but brittle	very hard (in case of three-dimensional network structure)	solids usually soft	hard, malleable and ductile
Melting point and boiling point	high	very high	low	moderately high to high
Solubility in water and non-aqueous solvents	usually soluble in water but insoluble in non-aqueous solvents	insoluble in water and non-aqueous solvents	usually insoluble in water but soluble in non-aqueous solvents	usually insoluble in water and non-aqueous solvents
Electrical conductivity	good conductors when molten or dissolved in water	non-conductors (except graphite)	non-conductors	good conductors

Some metals react with water to give hydrogen.

Have you mastered?

2 Allotropes are two (or more) forms of the same element in which the atoms or molecules are arranged in different ways.

3 a) Graphite is soft because the van der Waals' forces between the layers of carbon atoms are weak.

b) In graphite, the layers of carbon atoms can easily slide past each other. Thus, graphite is used as a lubricant.

c) Graphite is a good conductor of electricity because it contains delocalized electrons that are mobile.

4 From the group numbers of elements that make up a compound, we can tell

- the type of bonding present in the compound;
- the structure of the compound;
- the properties of the compound.

5 Properties such as melting point and electrical conductivity of a substance can help us to predict its bonding and structure.

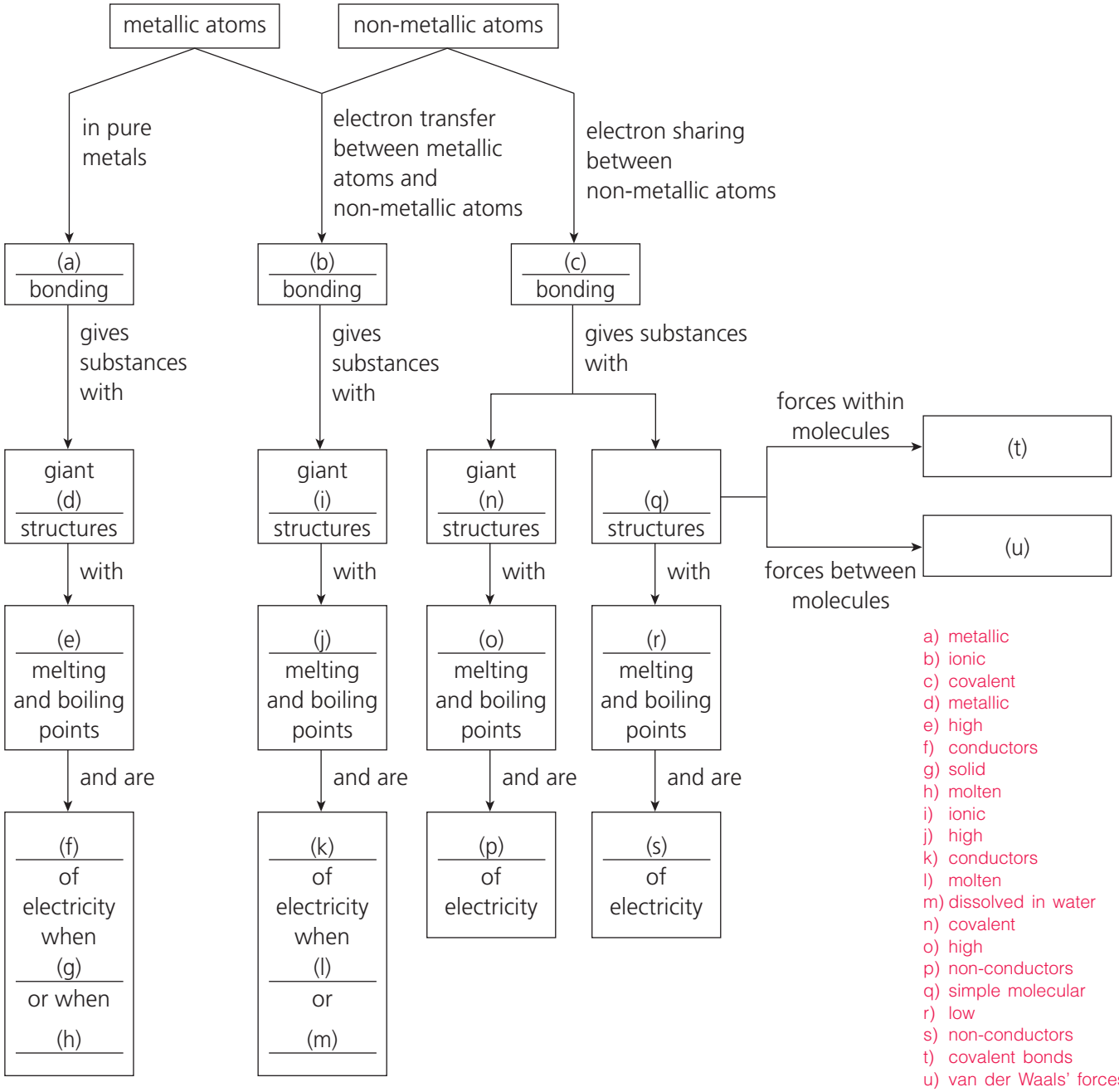
Unit Exercise

Note: The symbol ! indicates the level of difficulty of a question.

Part I Knowledge and understanding

1 Complete the concept map using the following words:

covalent (2 times), conductors (2 times), covalent bonds, dissolved in water, high (3 times), ionic (2 times), low, metallic (2 times), molten (2 times), non-conductors (2 times), simple molecular, solid, van der Waals' forces



a) metallic
b) ionic
c) covalent
d) metallic
e) high
f) conductors
g) solid
h) molten
i) ionic
j) high
k) conductors
l) molten
m) dissolved in water
n) covalent
o) high
p) non-conductors
q) simple molecular
r) low
s) non-conductors
t) covalent bonds
u) van der Waals' forces

2 Arrange the substances below in four groups, according to their structures.

aluminium	potassium chloride	water	quartz	potassium
calcium carbonate	diamond	sugar	tetrachloromethane (CCl_4)	oxygen

Giant ionic structure	Giant covalent structure	Simple molecular structure	Giant metallic structure
potassium chloride calcium carbonate	quartz diamond	water sugar tetrachloromethane oxygen	aluminium potassium

3 When some compounds are melted, they will then conduct electricity.

Draw ONE straight line to join the two statements which BEST explain why this is so.

Particles (choose one)	**Behaviour** (choose one)
The particles in each element are ions.	In a melted compound, electrons are passed from ion to ion.
The particles in the compound make up a regular lattice.	In a melted compound, the ions can move.
The particles in the compound are ions.	Particles in a lattice vibrate more at higher temperatures.

(OCR GCSE 21st Century Science (Higher Tier), Additional Science A, Unit 1, Jun. 2008, 4(b))

4 Complete the following statements about graphite and diamond.

Graphite is soft because it consists of layers of atoms held together in hexagonal arrangement by ___covalent___ bonds but with ___weak van der Waals' forces___ between the layers. The layers are able to ___slide over each other___ and this enables graphite to act as a lubricant. It is a conductor of electricity because it has ___delocalized electrons that are mobile / can move___ .

Each carbon atom in diamond has four covalent bonds arranged in a ___tetrahedral___ shape around the carbon and this makes the crystal very ___hard___ . It does not conduct electricity because all four outermost shell electrons of each carbon atom are ___used in bonding___ .

(Edexcel GCE O Level, Paper 1, Jan. 2010, 6)

5 Joe finds some information about the bonding in different chemicals.

a) The table shows some of the information he finds.

Complete the table by putting a tick '✔' in one box in each row to show the type of bonding in each chemical.

Chemical	Conducts electricity when solid	Conducts electricity when molten	Ionic	Covalent	Metallic
A	yes	yes			✔
B	no	yes	✔		
C	no	no		✔	

b) Joe writes down some general statements about ionic and covalent compounds.

Put a tick '✔' in one box in each row to show whether each statement is only true for ionic compounds or only true for covalent compounds or true for both.

	Only true for ionic compounds	Only true for covalent compounds	True for both
Conduct electricity when dissolved in water	✔		
May have melting and boiling points below room temperature		✔	
May have weak forces of attraction between molecules		✔	
Atoms are held together by forces between nuclei and shared electrons		✔	
May be solids at room temperature			✔

(OCR GCSE 21st Century Science (Higher Tier), Chem. A, Unit 2, Jan. 2012, 3)

Part II Multiple choice questions

6 Which of the substances listed in the following table has a giant ionic structure?

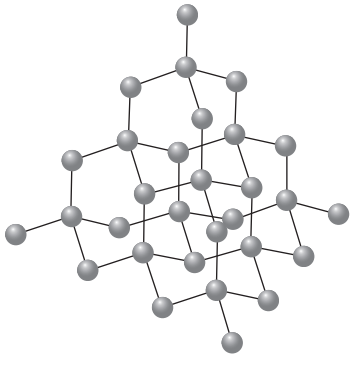
Substance	Electrical conductivity in solid state	Electrical conductivity in molten state
W	not conducting	not conducting
X	not conducting	conducting
Y	conducting	not conducting
Z	conducting	conducting

A W
(B) X
C Y
D Z

(HKCEE, Paper 2, 2010, 17)

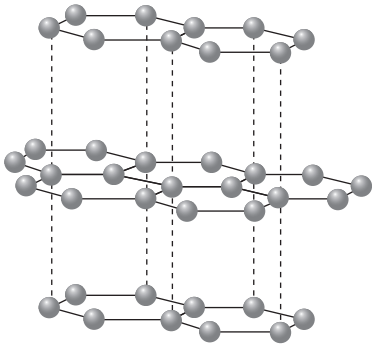
Directions: Use this information to answer questions 7 and 8.

The diagrams below show the structures of diamond and graphite.



key:
carbon atom

diamond



graphite

7 Diamond is very hard because

A ionic bonds are very strong.
B it has strong intermolecular forces.
(C) each carbon atom in its structure forms four strong covalent bonds.
D all forms of carbon are very hard.

(Edexcel GCSE (Foundation and Higher Tiers), Chemistry, Unit C2, Nov. 2011, 19)

8 Graphite is a good conductor of electricity because

A all forms of carbon are good conductors of electricity.
B every carbon atom in its structure forms a carbon ion.
(C) each carbon atom in its structure is bonded to only three others, resulting in delocalized electrons.
D the weak forces between the layers in the structure allow atoms to move easily.

(Edexcel GCSE (Foundation and Higher Tiers), Chemistry, Unit C2, Nov. 2011, 20)

9 Which of the following statements about silicon dioxide is correct?

A It consists of discrete molecules.
B It melts upon heating in a test tube.
C It is ductile.
(D) It is a poor conductor of electricity.

(HKDSE, Practice paper 1A, 2012, 4)

10 At room temperature, chlorine is a gas and bromine is a liquid. This is because

A bromine molecules are more stable than chlorine molecules.
B bromine atoms are smaller than chlorine atoms with stronger forces between the atoms in bromine molecules.
(C) the forces of attraction between bromine molecules are larger than the forces of attraction between chlorine molecules.
D bromine atoms contain fewer electrons than chlorine atoms.

(Edexcel GCSE (Higher Tier), Chemistry, Unit C2, Jun. 2012, 26)

Van der Waals' forces between bromine molecules are stronger than those between chlorine molecules.

11 Strong covalent bonds exist between atoms in sulphur dioxide molecules.
Weak van der Waals' forces exist between sulphur dioxide molecules.

11 Sulphur dioxide exists as molecules with the formula SO_2.

The table below shows the strength of the bonds between atoms in sulphur dioxide molecules and the strength of the forces between the molecules.

Which row of the table is correct?

	Strength of bonds between atoms in sulphur dioxide molecules	Strength of forces between sulphur dioxide molecules
A	weak	strong
B	weak	weak
(C)	strong	weak
D	strong	strong

(Edexcel GCSE (Foundation and Higher Tiers), Chemistry, Unit C2, Jun. 2012, 22)

12 Which of the following properties is NOT possessed by both carbon and nitrogen?

A They can form multiple bonds.
(B) They can exist in giant covalent structures.
C They are chemically stable at room temperature and pressure.
D They react with oxygen under suitable conditions to form acidic oxides.

(HKCEE, Paper 2, 2009, 3)

13 Metals usually have high melting points and boiling points because there are

A strong attractions between the ions.
B strong attractions between the delocalized electrons.
(C) strong attractions between the ions and the delocalized electrons.
D strong intermolecular forces.

Strong metallic bonds exist in metals.

(Edexcel Advanced Subsidiary GCE, Unit 1, Jan. 2011, 12)

14 Which of the following substances CANNOT conduct electricity?

A Pt(s)
B $PbBr_2(l)$
C C(graphite)
(D) $CH_3CH_2OH(l)$

(HKDSE, Paper 1A, 2012, 1)

15 Consider the information below:

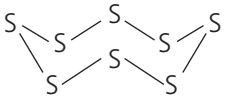
Solid	Melting point	Electrical conductivity	Solubility in water
W	low	good	insoluble
X	high	non-conducting	soluble
Y	low	non-conducting	insoluble
Z	high	non-conducting	insoluble

Which of the following solids is likely to be a covalent compound with a simple molecular structure?

A W
B X
(C) Y
D Z

A covalent compound with a simple molecular structure has a low melting point. It cannot conduct electricity.

16 The structure of a sulphur molecule in sulphur powder is shown below:



Which of the following bonds or attractive forces exist in sulphur powder?

(1) Covalent bond
(2) Ionic bond
(3) Van der Waals' forces

A (1) only
B (2) only
(C) (1) and (3) only
D (2) and (3) only

(3) Van der Waals' forces exist between sulphur molecules.

17 A substance has a high melting point and can conduct electricity when molten. It may be

(1) aluminium.
(2) phosphorus.
(3) magnesium chloride.

A (1) only
B (2) only
(C) (1) and (3) only
D (2) and (3) only

(1) Aluminium conducts electricity when in solid and molten states.
(2) Phosphorus does not conduct electricity.
(3) Magnesium chloride does not conduct electricity in the solid state. However, it conducts electricity when molten.

Unit Exercise

Part III Structured questions

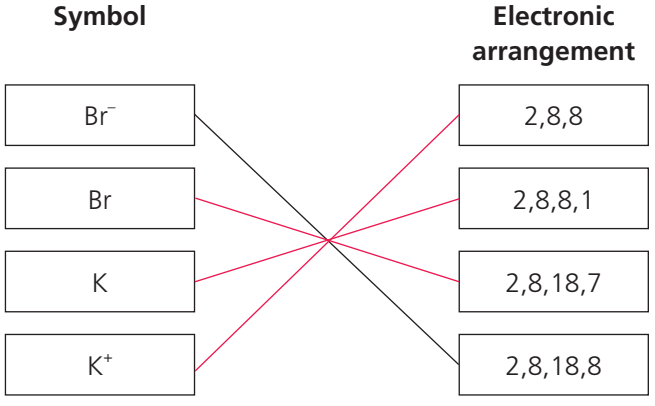
Answers for the questions in Hong Kong public examinations are not provided.

18 Potassium bromide contains potassium ions, K^+, and bromide ions, Br^-.

If an atom gains an electron, it turns into a negative ion. If an atom loses an electron, it turns into a positive ion.

a) Draw lines to link each symbol for an atom or ion to its correct electronic arrangement.

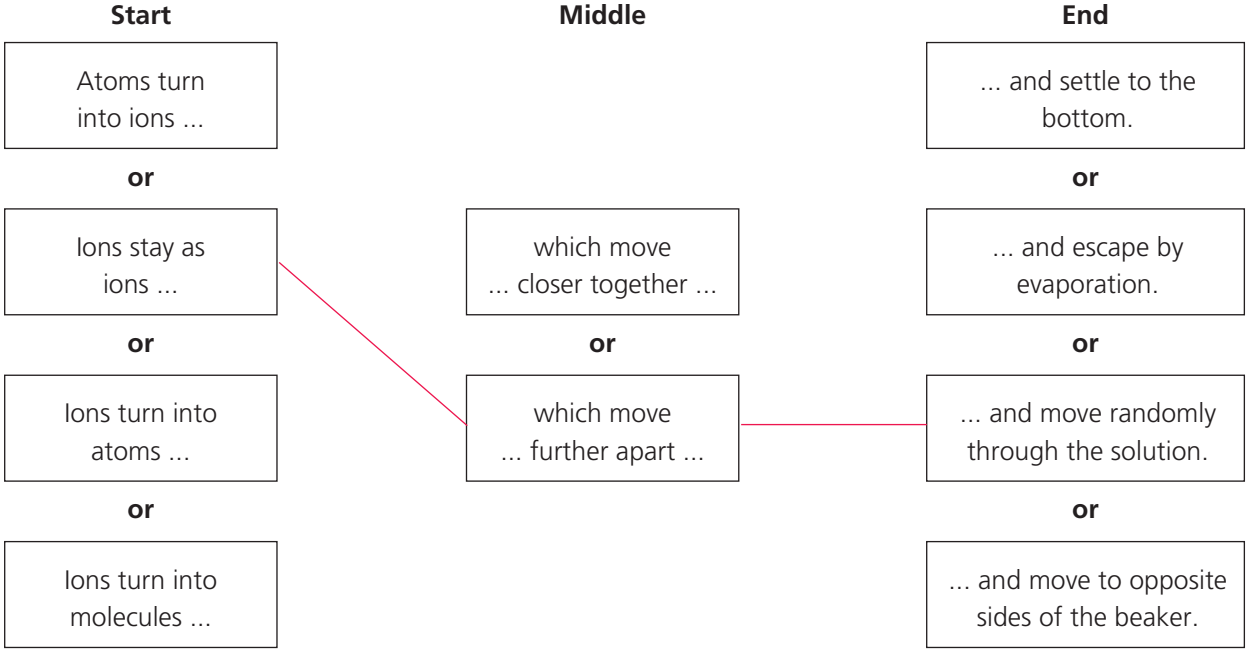
One has been done for you.



b) The potassium bromide forms ionic crystals.

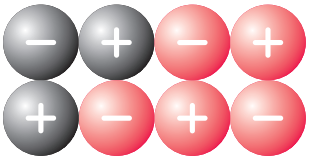
Matilda dissolves potassium bromide crystals in a beaker of water.

Complete ONE sentence from the choices below to describe what happens to potassium bromide crystals when they dissolve.



c) Matilda starts to draw a two-dimensional diagram of the ions in a potassium bromide crystal.

Continue her diagram by drawing in FOUR more of the ions.



d) Matilda does an experiment to show that potassium bromide is ionic.

How does she do this?

Measure the electrical conductivity of molten potassium bromide or potassium bromide dissolved in water.

Your answer should include

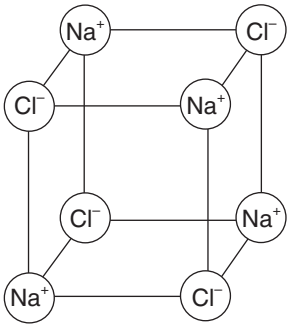
Ionic compound potassium bromide can conduct electricity.

- what she should do; and
- how she can tell that the potassium bromide is ionic.

(OCR GCSE 21st Century Science (Higher Tier), Additional Science A, Unit 1, May 2011, 18)

19 Sodium chloride, NaCl, and magnesium oxide, MgO, are both ionic compounds.

a) Look at the diagram. It shows the giant ionic structure of sodium chloride.



Sodium chloride contains sodium ions, Na^+, and chloride ions, Cl^-.

i) Sodium ions are made from sodium atoms. Describe how.

ii) Solid sodium chloride does NOT conduct electricity. Explain why.

iii) Sodium chloride solution does conduct electricity. Explain why.

b) Draw an electron diagram for magnesium oxide.

The electronic arrangement for a magnesium atom is 2,8,2 and that for an oxygen atom is 2,6.

Include in your answer:

- the electronic structure of the ions formed; and
- the charges on the ions.

(OCR GCSE Gateway Science, Chem. B, Unit 1, Jun. 2009, 10)

20 Both caesium (Cs) and sodium (Na) are elements in Group I of the periodic table. Caesium reacts with chlorine to form caesium chloride.

a) Solid caesium chloride has a giant ionic structure.

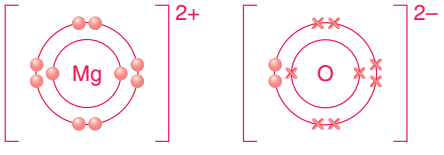
i) Draw a diagram to show the structure of caesium chloride.

ii) Explain why solid caesium chloride is brittle.

b) Predict, with ONE reason, whether sodium or caesium is more reactive towards chlorine.

(HKDSE, Paper 1B, 2013, 8(b)(i)–(c))

19 a) i) Loss of electron(s)
ii) The ions are in fixed positions.
iii) The ions can move in a solution.

b) 

Unit Exercise

21 This question is about the properties of ions and ionic compounds. Refer to page T45.

a) Solid calcium carbonate, $CaCO_3$, has a giant ionic structure.

i) Draw a diagram for a calcium ion. Show ALL the electrons and the charge on the ion.

ii) Explain why ionic compounds have relatively high melting points.

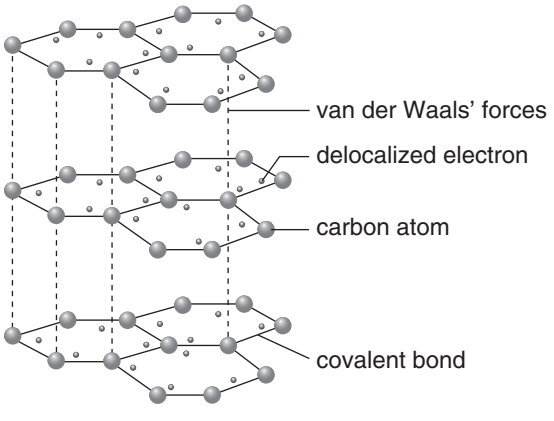
b) Changes in the concentration of ions in a solution can be estimated by measuring the electrical conductivity of the solution.

i) Explain why solutions of ions are able to conduct electricity.

ii) Suggest why aqueous solutions of calcium chloride, $CaCl_2(aq)$, and barium chloride, $BaCl_2(aq)$, of the same concentration, have different electrical conductivities.

(Edexcel Advanced Subsidiary GCE, Unit 1, May 2010, 15(a)(i), (iv), (b)(i)–(ii))

22 Graphite is a form of carbon and can be used to make pencil cores and electrodes. There are layers of carbon atoms in the structure of graphite. In each layer, each carbon atom is linked to other carbon atoms by covalent bonds. Moreover, delocalized electrons exist in the layers, while van der Waals' forces exist between the layers as shown in the diagram below:



a) Answer the following questions according to the information given above.

i) Explain why the graphite used to make pencil cores can be easily detached to form markings on paper.

ii) Explain why graphite can be used to make electrodes.

b) Lead metal can also be used to draw markings. With reference to the bonding of lead, explain why using lead to make pencil cores is not as good as using graphite.

c) Diamond is another form of carbon. With reference to the bonding and structure of diamond, explain why diamond is so hard.

(HKCEE, Paper 1, 2011, 3)

23 Soda-lime glass is made by heating, to above 1 500 °C, a mixture of soda (sodium carbonate), limestone and sand. Refer to page T45.

a) i) Suggest a test to show that soda contains sodium. State your expected observation.

ii) Explain why sodium carbonate is soluble in water.

b) i) Name the main chemical constituent of limestone.

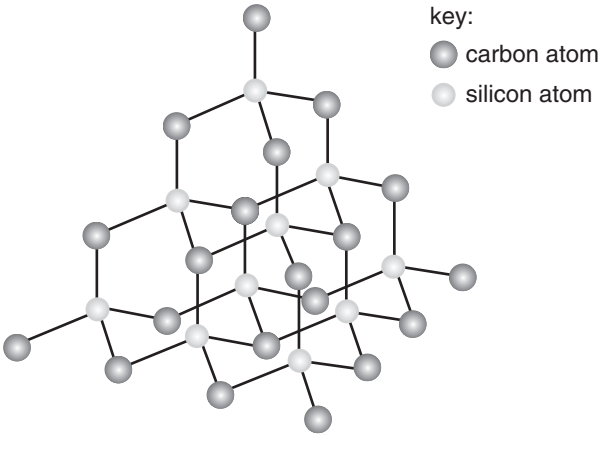
ii) Limestone breaks down when heated to above 1 500 °C. Write a word equation for the reaction involved.

c) i) Sand is an impure form of quartz. Draw a three-dimensional structure of quartz.

ii) Explain why quartz is stable in moist air.

24 Silicon carbide, SiC, is used in cutting tools.

Look at the diagram. It shows the structure of silicon carbide.



silicon carbide

a) Silicon carbide has a giant covalent structure.
To melt it, a lot of heat is needed to break the strong covalent bonds between the atoms.

Silicon carbide and diamond have similar structures.

a) Suggest why silicon carbide has a high melting point.

Use ideas about the structure of silicon carbide.

b) One reason for silicon carbide being used in cutting tools is that it has a high melting point.

Suggest one other reason why silicon carbide is used in cutting tools. Very hard

(OCR GCSE Gateway Science, Chem. B, Unit 2, Jan. 2011, 4)

25 Scientists have recently developed a method to produce large sheets of a substance called graphene. Graphene is made from carbon and is a single layer of graphite just one atom thick. Refer to page T45.

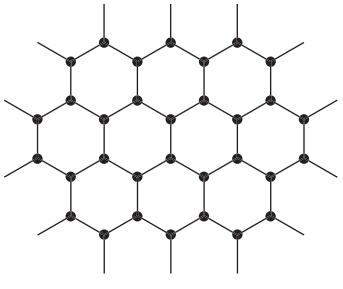
The properties of graphene include:

- it conducts electricity;
- it is transparent since it is only one atom thick; and
- it is strong and durable.

These properties make it suitable to overlay a monitor screen to make it a touchscreen.



The diagram below shows the structure of graphene.



a) Use your knowledge of the bonding in graphite and the diagram of the structure to help you to explain, as fully as you can:

i) why graphene is strong; and

ii) why graphene conducts electricity.

b) Suggest why a sheet of graphite which has a large number of carbon layers would not be suitable for the touchscreen.

(AQA GCSE (Higher Tier), Chemistry, Unit 2, Jan. 2012, 7)

26 a) i) Draw an electron diagram to show the arrangement of the electrons in a molecule of ammonia, NH_3. Refer to page T46.

Show *outermost shell* electrons only.

ii) Give the name of the type of bond between nitrogen and hydrogen atoms in an ammonia molecule.

b) Ammonia has a simple molecular structure.

It is a gas at room temperature. The boiling point of liquid ammonia is –34 °C.

Explain why liquid ammonia has a low boiling point.

(Edexcel GCSE (Higher Tier), Chemistry, Unit C2, Nov. 2009, 3(c)–(d))

27 Calcium fluoride is an ionic compound. It can be made by reacting calcium with fluorine. Refer to page T46.

a) Calcium fluoride consists of calcium ions, Ca^{2+}, and fluoride ions, F^-.

i) Explain how a calcium atom, Ca, forms a calcium ion, Ca^{2+}.

ii) Write the chemical formula of calcium fluoride.

b) Explain why calcium fluoride has a high melting point.

c) When molten calcium fluoride is electrolyzed, fluorine gas is formed at the positive electrode.

Name the element formed at the negative electrode.

d) The element fluorine exists as fluorine molecules, F_2.

i) Draw an electron diagram to show the arrangement of electrons in one molecule of fluorine.

Show *outermost shell* electrons only.

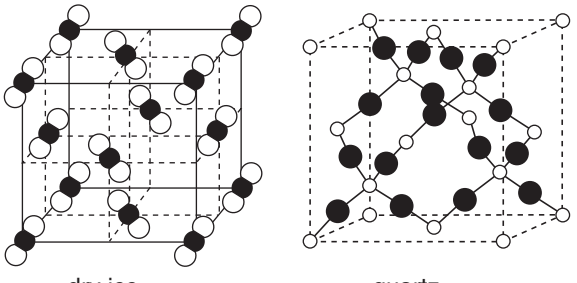
ii) Fluorine can be made into a solid by cooling it.

Solid fluorine melts at –220 °C.

Explain why fluorine has such a low melting point.

(Edexcel GCSE (Higher Tier), Chemistry, Unit C2, Mar. 2010, 4)

28 Both carbon and silicon are Group IV elements in the periodic table. The diagrams below show the structures of dry ice (solid carbon dioxide) and quartz (a form of silicon dioxide): Refer to page T46.



dry ice quartz

a) Explain why carbon dioxide is a gas at room temperature and pressure, whereas silicon dioxide is a solid with a very high melting temperature.

b) The structure of silicon is similar to that of diamond.

i) Draw the three-dimensional structure of silicon.

ii) Explain whether or not you would expect silicon to have a high melting point.

c) Draw an electron diagram of silicon tetrachloride, showing electrons in the *outermost shells* only.

32 a) The delocalized electrons are mobile.
b) The layers of positively charged metallic ions slide over each other when the metal is hammered.
New metallic bonds are re-formed.

29 Iodine and graphite are both solids. When iodine is heated gently, a purple vapour is seen. Graphite will not melt until the temperature reaches 4 000 K. Graphite conducts electricity but iodine is a very poor conductor of electricity. Refer to page T47.

a) State the type of crystal structure for each of iodine and graphite.

b) Describe the structure of and bonding in graphite and explain why the melting point of graphite is very high.

c) Explain why iodine vaporizes when heated gently.

d) State why iodine is a very poor conductor of electricity.

(AQA Advanced Subsidiary GCE, Unit 1, Jun. 2010, 7)

30 Explain why iodine is slightly soluble in water but very soluble in non-aqueous solvents. Refer to page T47.

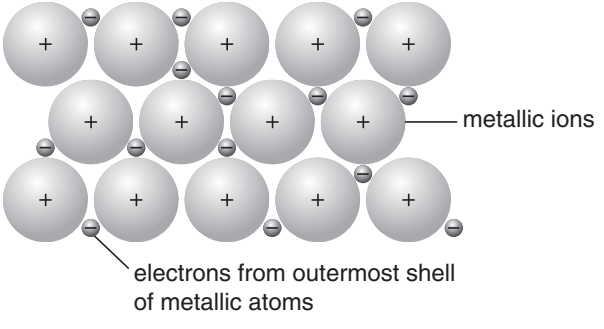
31 Explain each of the following. Refer to page T47.

a) Sodium metal conducts electricity in both the solid and molten states whereas sodium chloride only conducts when molten.

b) Many covalent compounds are gases or liquids whereas ionic compounds are solids at room temperature.

(Edexcel GCE O Level, Paper 2, Jan. 2009, 8(a))

32 The diagram represents the structure of a metal.



Use the diagram to help you explain

a) why a metal conducts electricity; and

b) why a metal is malleable.

(Edexcel IGCSE, Chemistry, Paper 2, Nov. 2008, 6(a))

33 This question is about elements in Period 2 of the periodic table. Refer to page T48.

a) Lithium has a giant metallic structure and a boiling point of 1 342 °C.

Describe, with the aid of a labelled diagram, the structure and bonding in lithium and explain why lithium has a high boiling point.

b) Fluorine is a gas at room temperature and has a very low boiling point of –188 °C.

i) Draw an electron diagram to show the bonding in a fluorine molecule. Show *outermost shell* electrons only.

ii) Explain why fluorine has a low boiling point.

c) Fluorine reacts with lithium at room temperature to form a white crystalline solid, lithium fluoride. Lithium fluoride is a good conductor of electricity when molten but NOT when solid.

i) Draw an electron diagram to show the bonding in lithium fluoride. Show the *outermost shell* electrons only.

ii) Explain why lithium fluoride conducts electricity when molten but NOT when solid.

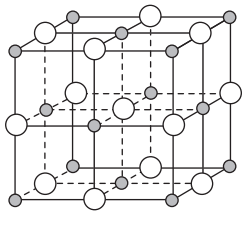
(OCR Advanced Subsidiary GCE, Chem. A, F321, May 2010, 5(a)–(c))

34 Consider the substances listed below:

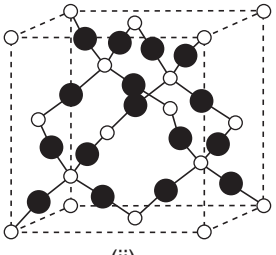
argon, hydrogen chloride, magnesium oxide, oxygen, silicon dioxide, sulphur dioxide

Diagrams (i), (ii), (iii) and (iv) below show the arrangements of particles, which can be atoms, ions or molecules, in some substances.

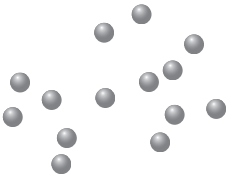
a) For each diagram, choose ONE substance from the above list, which has the arrangement of particles as shown under room temperature and pressure.



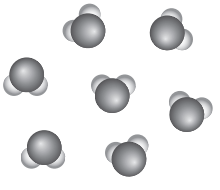
(i)



(ii)



(iii)



(iv)

b) Use tick(s) '✔' to show the chemical bond and / or attractive forces existing in each substance suggested in (a).

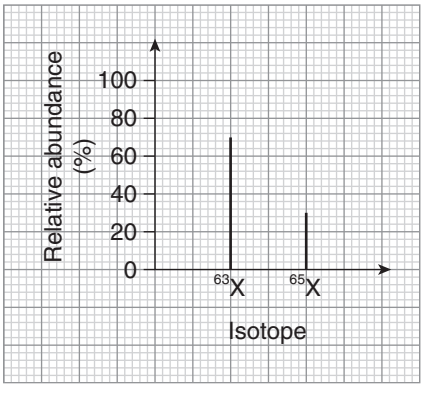
Diagram	Substance	Chemical bond and / or attractive forces existing in the substance		
		Covalent bond	Ionic bond	Van der Waals' forces
(i)	magnesium oxide		✔	
(ii)	silicon dioxide	✔		
(iii)	argon			✔
(iv)	sulphur dioxide	✔		✔

Topic Exercise

Note: The symbol ʃ indicates the level of difficulty of a question.

Part I Multiple choice questions

1 Suppose that element X has only two isotopes, ^{63}X and ^{65}X. The graph below shows the relative abundance of the two isotopes.



Which of the following is the relative atomic mass of X?

A 63.3
B 63.5
C 63.6
D 64.0

(HKDSE, Paper 1A, 2013, 2)

ʃ2 Naturally occurring magnesium has three isotopes: ^{24}Mg, ^{25}Mg and ^{26}Mg. The relative abundance of the ^{25}Mg isotope is 10%. What is the relative abundance of the ^{26}Mg isotope?

(Relative atomic mass: Mg = 24.3)

A 10%
B 15%
C 23%
D 85%

(HKCEE, Paper 2, 2010, 8)

3 X^{2+} ion has an electronic arrangement of 2,8,8. Which of the following statements concerning the carbonate of X is INCORRECT?

A It is a white solid.
B It is soluble in water.
C It decomposes on heating.
D It produces a brick red flame in flame test.

(HKCEE, Paper 2, 2010, 3)

ʃʃ4 X, Y and Z are three different elements. X^{2+} ion, Y^- ion and argon atom have the same electronic arrangement. Z belongs to the same period as X and the same group as Y in the periodic table. What is Z?

A Mg
B Cl
C Ca
D Br

(HKCEE, Paper 2, 2011, 31)

ʃ5 The electronic arrangements of three chemical species are shown below.

W : 2,1
X^{2+} : 2,8
Y^- : 2,8,7

Which of the elements, W, X and Y, are in the same period of the periodic table?

A W and X only
B W and Y only
C X and Y only
D W, X and Y

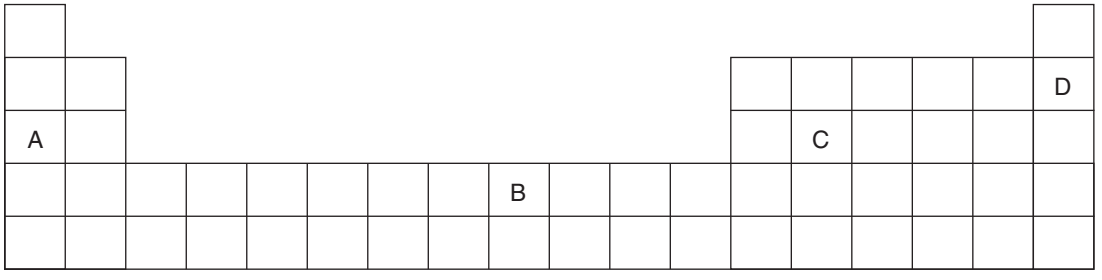
(HKCEE, Paper 2, 2009, 1)

6 Both radium (Ra) and calcium (Ca) belong to the same group of the periodic table. Which of the following statements is INCORRECT?

A Radium is a good conductor of electricity in the solid state.
B Radium atoms readily donate electrons to form Ra^{2+} ions.
C Both radium and calcium become tarnished after exposed to air for some time.
(D) Radium is less reactive than calcium.

(HKDSE, Paper 1A, 2013, 12)

Direction: Questions 7 and 8 refer to the following outline of the periodic table. The letters A to D are NOT the symbols of the elements.



7 Select from A to D the element which is a non-metal with a high melting point and boiling point.

C
Element C is silicon. It has a giant covalent structure. Thus, it has a high melting point and boiling point.

(Edexcel Advanced Subsidiary GCE, Unit 1, May 2011, 1(a))

8 Select from A to D the element which has a very stable electronic structure.

D
Element D is neon. Its atom has a very stable electronic structure (2,8).

(Edexcel Advanced Subsidiary GCE, Unit 1, May 2011, 1(c))

9 Silicon is an element in Group IV of the periodic table. The oxide of silicon has the chemical formula SiO_2. Which of the following statements about silicon and its oxide is correct?

A Silicon is a good conductor of heat.
B Silicon exists as simple molecules.
(C) SiO_2 is a hard material at room temperature.
D SiO_2 dissolves in water to give an acidic solution.

(HKDSE, Paper 1A, 2013, 1)

10 Which of the following species is NOT an appropriate example for illustrating dative bond formation?

(A) NH_3
B NH_4^+
C BF_4^-
D BF_3NH_3

(HKDSE, Practice paper 1A, 2012, 3)

11 Which of the following statements concerning an ^{131}I atom and a ^{131}Xe atom is / are correct?

(1) They have the same number of protons.
(2) They have different numbers of neutrons.
(3) They have different numbers of outermost shell electrons.

A (1) only
B (2) only
C (1) and (3) only
(D) (2) and (3) only

(HKDSE, Paper 1A, 2012, 15)

12 Comparing the elements in the second period of the periodic table, from lithium to fluorine, which of the following statements is / are correct?

(1) They show a gradual change from having metallic property to having non-metallic property.
(2) They show a gradual increase in the number of electron shells in their atoms.
(3) They show a gradual decrease in melting point.

(A) (1) only
B (2) only
C (1) and (3) only
D (2) and (3) only

(HKCEE, Paper 2, 2008, 22)

13 A substance has a high melting point and does NOT conduct electricity when in solid state. It may be

(1) a compound with giant molecules.
(2) an element with giant covalent structure.
(3) a compound with giant ionic structure.

A (1) and (2) only
B (1) and (3) only
C (2) and (3) only
(D) (1), (2) and (3)

(HKCEE, Paper 2, 2009, 22)

Directions : Each question (Questions 14–17) consists of two separate statements. Decide whether each of the two statements is true or false; if both are true, then decide whether or not the second statement is a correct explanation of the first statement. Then select one option from A to D according to the following table :

A Both statements are true and the 2nd statement is a correct explanation of the 1st statement.
B Both statements are true but the 2nd statement is NOT a correct explanation of the 1st statement.
C The 1st statement is false but the 2nd statement is true.
D Both statements are false.

	1st statement	2nd statement	
14	Isotopes of an element have the same mass.	Isotopes of an element have the same number of protons. *(HKCEE, Paper 2, 2007, 29)*	C
15	Ammonium chloride is an ionic compound.	Ammonium chloride can conduct electricity in aqueous state. *(HKCEE, Paper 2, 2009, 28)*	B
16	Copper(II) carbonate dissolves in water to give a blue solution.	All solid compounds with copper(II) as the only cations are blue in colour. *(HKCEE, Paper 2, 2009, 29)*	D
17	Molten sulphur is a good conductor of electricity.	Sulphur molecules are mobile in molten sulphur. *(HKCEE, Paper 2, 2007, 28)*	C

Part II Structured questions

Answers for the questions in Hong Kong public examinations are not provided.

18 Robert Bunsen, the inventor of the Bunsen burner, was partly responsible for the discovery of the element rubidium.

A sample of rubidium was analyzed and found to consist of two isotopes, rubidium-85 and rubidium-87. Information about these isotopes is given in the table.

Isotope	Relative isotopic mass	Relative abundance (%)
Rubidium-85	85.00	72.15
Rubidium-87	87.00	27.85

a) In terms of subatomic particles, what is the difference between these isotopes of rubidium?

b) Define the term 'relative atomic mass'.

c) Calculate the relative atomic mass of the sample of rubidium, giving your answer to TWO decimal places.

(OCR Advanced Subsidiary GCE, Chem. A, F321, Jan. 2011, 1(a)–(c))

19 Boron occurs naturally in two isotopes, ^{10}B and ^{11}B.

a) What is meant by the term 'isotopes'?

b) With reference to the periodic table, calculate the percentage abundance of ^{11}B in nature.

c) $^{10}BCl_3$ and $^{11}BCl_3$ are compounds formed respectively from the two isotopes of boron with chlorine. $^{10}BCl_3$ reacts with water to give white fumes. State, with explanation, the expected observation when $^{11}BCl_3$ is added to water.

(HKCEE, Paper 1, 2008, 2)

18 a) Rubidium-87 has two more neutrons.

b) The relative atomic mass of an element is the weighted average relative isotopic mass of all the naturally occurring isotopes of that element compared with $\frac{1}{12}$ of the mass of one carbon-12 atom.

c) Relative atomic mass of rubidium

$$= \frac{85.00 \times 72.15 + 87.00 \times 27.85}{100}$$

$$= 85.56$$

20 The following table shows the information about some Group II elements.

Element	Symbol	Atomic number	Electronic arrangement of atom	Melting point (°C)
Beryllium	Be	4	2,2	1 280
Magnesium	Mg	12	2,8,2	650
Calcium	Ca	20	2,8,8,2	838
Strontium	Sr	38	?	?
Barium	Ba	56	?	725

a) What is the electronic arrangement of a strontium atom?
2,8,18,8,2

b) Estimate the melting point of strontium.
In between of 725 °C and 838 °C

c) To which period of the periodic table does barium belong?
Period 6

d) Suggest why these Group II elements react with water in a similar way.
Atoms of these Group II elements have the same number

e) Which of the above elements is the most reactive?
Barium

f) Barium reacts with chlorine to form barium chloride.

i) Draw an electron diagram of barium chloride, showing electrons in the *outermost shells* only.
Refer to page T49.

ii) State the chemical formula of barium chloride.
$BaCl_2$

iii) Discuss, with explanation, the electrical conductivity of barium chloride with reference to the type and property of particles in it.
Solid barium chloride cannot conduct electricity.
Barium chloride conducts electricity when in molten state or when dissolved in water.
The ions in solid barium chloride are in fixed positions.
The ions become mobile when barium chloride becomes molten or dissolved in water.

21 Neon occurs naturally in three isotopes with the relative abundance of each isotope shown in the table below.

Isotope	Relative abundance (%)
^{20}Ne	90.48
^{21}Ne	0.27
^{22}Ne	9.25

a) What is meant by the term 'isotope'?

b) Calculate the relative atomic mass of neon.

c) Give one daily application of neon.

d) Explain why the boiling point of neon is lower than that of oxygen.

(HKDSE, Paper 1B, 2012, 1)

22 a) Many scientists contributed to the development of the periodic table. One of these was Dmitri Mendeléev. In his table, he arranged the elements in order of increasing relative atomic mass.

In the modern periodic table, elements are arranged in order of increasing atomic number.

i) Explain the term 'atomic number'.

ii) When the elements are arranged in order of increasing relative atomic mass, iodine would appear in Group VI.

Give ONE reason why iodine must appear in Group VII.

b) Gallium is the element below aluminium in the periodic table.

When Mendeléev produced his periodic table, gallium had not been discovered. He left a gap for gallium below aluminium and predicted the properties of gallium and its compounds.

The table below shows Mendeléev's predictions and the properties we now know gallium to have.

	Mendeleev's prediction	Properties of gallium
Relative atomic mass	about 68	70
Melting point (°C)	low	30
Reaction of oxide with acids	forms colourless solution	forms colourless solution

Give ONE piece of evidence from the table that shows that Mendeléev's predictions were correct.

c) Aluminium oxide and gallium oxide are ionic compounds.

i) The chemical formula of aluminium oxide is Al_2O_3. The symbol for an atom of gallium is Ga.

Suggest the formula of gallium oxide.

ii) Calculate the formula mass of aluminium oxide, Al_2O_3.

(Relative atomic masses: O = 16.0, Al = 27.0)

iii) Predict ONE property of gallium oxide that is NOT given in the table.

d) Mendeléev did not know of the existence of the noble gases.

Helium and neon are noble gases and do not form any compounds.

Why do helium and neon not form any compounds?

(Edexcel GCSE (Higher Tier), Chemistry, Unit C2, Nov. 2009, 1)

22 a) i) The atomic number of an element is the number of protons in an atom of that element.
ii) Any one of the following:
- It has 7 outermost shell electrons in its atom.
- Its chemical properties are similar to those of other Group VII elements.

b) Any one of the following:
- Similar relative atomic mass
- Low melting point
- The oxide forms colourless solution with acids

c) i) Ga_2O_3
ii) 2 x 27.0 + 3 x 16.0 = 102
iii) Any one of the following:
- Conducts electricity when molten / dissolved in water
- High melting / boiling point
- Any other property of ionic compounds

d) There are 2 electrons (helium) / 8 electrons (neon) in the outermost shell of the atom.

23 Both bromine (Br) and chlorine (Cl) are Group VII elements in the periodic table.

a) What is the name commonly given to this group of elements?

b) The electronic arrangement of bromine is 2,8,*p*,*q*.

p is _____ , *q* is _____ .

c) Explain, in terms of bonding and structure, why the boiling point of bromine is higher than that of chlorine.

d) Rubidium (Rb) is a Group I element in the periodic table. It reacts with bromine to form an ionic compound.

Write the electronic arrangement of a rubidium ion.

(HKCEE, Paper 1, 2010, 1(a)–(c), (d)(ii))

24 a) i) Define the term 'relative atomic mass' of an element.

Refer to page T49.

ii) A sample of the metal silver has the relative atomic mass of 107.9 and exists as two isotopes. In this sample, 54.0% of the silver atoms are one isotope with a relative isotopic mass of 107.1.

Calculate the relative isotopic mass of the other silver isotope. 108.8

State why the isotopes of silver have identical chemical properties.

b) Silver reacts with fluorine to form silver fluoride (AgF).

Silver fluoride has a high melting point and has a structure similar to that of sodium chloride.

i) State the type of bonding involved in silver fluoride.

ii) Draw a diagram to show how the particles are arranged in a silver fluoride lattice and show the charges on the particles.

iii) Explain why the melting point of silver fluoride is high.

(AQA Advanced Subsidiary GCE, Unit 1, Jun. 2009, 5(a), (d))

25 T, X and Z are three elements in the periodic table, with the sum of their atomic numbers equal to 38. Moreover, both T and X are Group VII elements, while the atomic number of T is smaller than that of X.

a) What are elements T, X and Z?

b) Draw the electron diagram of the compound formed from T and X, showing electrons in the *outermost shells* only.

c) Discuss, with explanation, the electrical conductivity of the compound formed from X and Z with reference to the type and property of the particles in it.

(HKCEE, Paper 1, 2008, 1)

26 Phosphorus is in the same group of the periodic table as nitrogen. A molecule of PH_3 reacts with an H^+ ion to form a PH_4^+ ion.

Name the type of bond formed when PH_3 reacts with H^+ and explain how this bond is formed.

(AQA Advanced Subsidiary GCE, Unit 1, Jan. 2009, 2(d))

27 BF_3 is a covalent molecule that reacts with an F^- ion to form a BF_4^- ion.

Name the type of bond formed when a molecule of BF_3 reacts with an F^- ion. Explain how this bond is formed.

(AQA Advanced Subsidiary GCE, Unit 1, Jun. 2010, 1(d)(i))

26 Type of bond: Dative covalent bond
Explanation: The lone pair on the phosphorus atom provides both the electrons for bonding with the hydrogen ion.

27 Type of bond: Dative covalent bond
Explanation: The fluorine ion provides both the electrons when bonded with the boron atom of BF_3.

28 Carbon occurs in a wide range of compounds and is essential to living systems.

Refer to page T50.

a) Two isotopes of carbon are ^{12}C and ^{13}C.

i) State what is meant by the term 'isotopes'.

ii) Isotopes of carbon have the same chemical properties. Explain why.

iii) The ^{12}C isotope is used as the standard measurement of relative masses.

Define the term 'relative isotopic mass'.

b) One form of naturally occurring carbon is graphite.

The table below lists some properties of graphite.

Electrical conductivity	good conductor
Hardness	soft
Melting point	very high

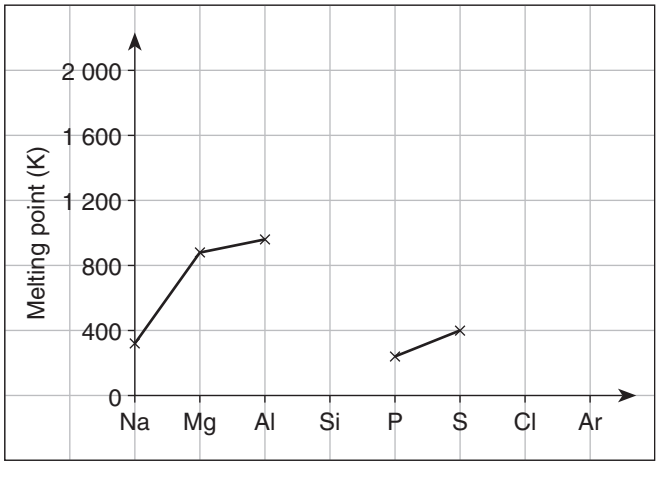
i) Describe the bonding and structure in graphite.

ii) Explain, in terms of bonding and structure, the properties of graphite shown above.

(OCR Advanced Subsidiary GCE, Chem. A, F321, Jan. 2009, 1(a)–(b))

29 The diagram below shows the melting points of elements in Period 3 of the periodic table.

Refer to page T50.



a) On the diagram, use crosses to mark the approximate positions of the melting points for the elements silicon, chlorine and argon. Complete the diagram by joining the crosses.

b) Draw a diagram to show the structure of silicon.

c) By referring to its structure and bonding, explain your choice of position for the melting point of silicon.

d) Suggest why chlorine is very soluble in non-aqueous solvents.

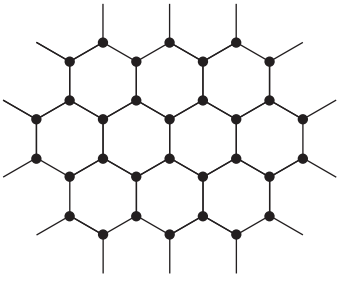
30 Use your knowledge of structure and bonding to deduce why sodium metal can be beaten into shape when hit with a hammer but solid sodium chloride breaks into small pieces.

(AQA Advanced Subsidiary GCE, Unit 1, Jun. 2009, 5(b))

31 a) Graphene is a new material made from carbon atoms. It is the thinnest and strongest material known. Graphene has a very high melting point and is an excellent conductor of electricity.

Refer to page T51.

Part of the structure of graphene is illustrated in the diagram.



i) Deduce the type of crystal structure shown by graphene.

ii) Suggest why graphene is an excellent conductor of electricity.

iii) Explain, in terms of its structure and bonding, why graphene has a high melting point.

b) Titanium is also a strong material that has a high melting point. It has a structure similar to that of magnesium.

i) State the type of crystal structure shown by titanium.

ii) Explain, in terms of its structure and bonding, why titanium has a high melting point.

30 When sodium metal is beaten, the layers of positive metallic ions slide over each other. New metallic bonds are re-formed. When solid sodium chloride is beaten, the ions may move so that ions with the same charge come together. Repulsion between thousands of ions in the layers causes the solid to break.

c) Titanium can be hammered into objects with different shapes that have similar strengths.

i) Suggest why titanium can be hammered into different shapes.

ii) Suggest why these objects with different shapes have similar strengths.

d) Magnesium oxide (MgO) has a melting point of 3 125 K.

Predict the type of crystal structure in magnesium oxide and suggest why its melting point is high.

(AQA Advanced Subsidiary GCE, Unit 1, May 2012, 4)

32 Solids exist as lattice structures.

Refer to page T51.

a) Giant metallic lattices conduct electricity. Giant ionic lattices do not. If a giant ionic lattice is melted, the molten ionic compound will conduct electricity.

Explain these observations in terms of bonding, structure and particles present.

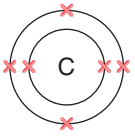
b) Solid SiO_2 melts at 2 230 °C. Solid $SiCl_4$ melts at –70 °C. Neither of the liquids formed conducts electricity.

Suggest the type of lattice structure in solid SiO_2 and in solid $SiCl_4$ and explain the difference in melting points in terms of bonding and structure.

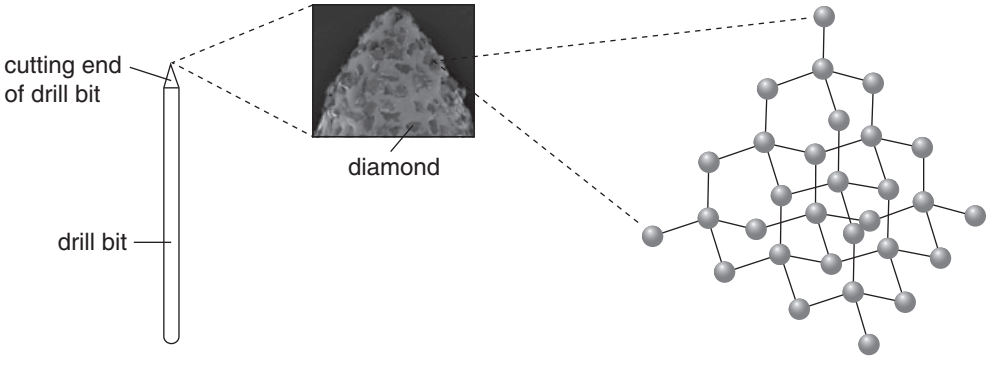
(OCR Advanced Subsidiary GCE, Chem. A, F321, Jan. 2011, 5(a), (c))

33 Pure carbon can exist in two forms, diamond and graphite. A carbon atom has 6 electrons.

a) Complete the diagram to show the electronic structure of a carbon atom. Show the electrons as crosses 'x'.



b) A drill bit is used to cut holes through materials. The cutting end of this drill bit is covered with very small diamonds.



the arrangement of atoms in diamond

i) What property of diamond makes it suitable for use on the cutting end of a drill bit?

ii) Explain, as fully as you can, why diamond has this property. Use your knowledge of the structure and bonding of diamond and the information shown above to help you to answer this question.

c) Explain why graphite is a good conductor of electricity and why diamond does NOT conduct electricity.

(AQA GCSE (Higher Tier), Chemistry, Unit 2, Jun. 2010, 3)

33 b) i) Hard
ii) Diamond has a giant covalent structure.
The strong covalent bonds restrict the relative motion of the atoms.
c) Graphite has delocalized electrons.
The electrons can more throughout the structure.
In diamond, the outermost shell electrons of each carbon atom are used for covalent bonding. There are no mobile electrons or ions.